MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, INTERFACE, DIGITAL, KEY CODE EKB

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic digital interface driver.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name	Driver, interface
Family	Digital
Key code	EKB
Size	1A
Weight	35 grams maximum
Failure rate	0.759 failures/10^6 hours maximum
Environment	Class I as specified in MIL-M-28787
Supply voltage	5.0 Vdc ±10 percent
V_{CC} input current (power dissipation)	
at V_{CC} = 5.5 V	
V_{CC} applied to pin 1 and pin 21	
Output at high level	88 mA maximum (484 mW)
Output at low level	110 mA maximum (605 mW)
V_{CC} applied to pin 1 or pin 21	
Output at high level	44 mA maximum (242 mW)
Output at low level	55 mA maximum (302 mW)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Input power dissipation (typical per circuit)	
A/C (V_{IH} = 28 V)	102 mW
B/C (V_{IH} = 5 V)	26 mW
E/C (V_{IH} = 115 Vrms, 400 HZ)	280 mW
Maximum power dissipation	
at V_{CC} = 5.5 V	
Output at high level	2.05 W
7 circuits A/C, V_{IH} = 35 V	
1 circuit E/C, V_{IH} = 127.5 Vrms	
Output at low level	0.605 W
8 circuits, V_{IL} = 0 V	
Differential low-level input voltage (V_{IL}) for a V_{OL}	
A/C (28 V input)	1.4 V maximum
B/C (5 V input)	0.7 V maximum
E/C (115 Vrms, 400 Hz input)	3.5 Vrms maximum 4.9 V_{pk} maximum
Differential low-level input current (I_{IL}) for a V_{OL}	
A/C (28 V input)	100 uA maximum
B/C (5 V input)	100 uA maximum
E/C (115 Vrms, 400 Hz input)	100 uA_{pk} maximum
Differential low-level input voltage (V_{IL}) at I_{IL}	
B/C I_{IL} at 100 uA	0.7 V minimum, 1.4 V maximum
Differential high-level input voltage (V_{IH}) for a V_{OH}	
A/C (28 V input)	20 V minimum, 35 V maximum
B/C (5 V input)	3.4 V minimum, 7.5 V maximum
E/C (115 Vrms, 400 Hz input)	103.5 Vrms minimum, 126.5 Vrms maximum
Differential high-level input current (I_{IH}) for a V_{OH}	
A/C (28 V input)	2.3 mA minimum
B/C (5 V input)	2.3 mA minimum
E/C (115 Vrms, 400 Hz input)	2.0 mA_{rms} minimum
Maximum differential input current (I_{IH}) at V_{IH} maximum	
A/C (V_{IH} = 35 V)	4.12 mA minimum, 5.13 mA maximum
B/C (V_{IH} = 7.5 V)	7.48 mA minimum, 10.45 mA maximum
E/C (V_{IH} = 126.5 Vdc)	2.37 mA minimum, 3.00 mA maximum

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Common mode voltage, input to output V_{ICR}	300 V maximum
Isolation resistance	1.0 MΩ minimum
Output voltage Low level (I_{OL} = 16 mA) High level (I_{OH} = -800 uA)	0.4 V maximum 2.4 V minimum
Output drive current maximum capability Low level (V_{OL} = 0.40 V) High level (V_{OH} = 2.40 V)	16 mA -800 uA
Output short circuit current I_{OS} (V_{CC} = 5.5 V)	-18 mA minimum, -55 mA maximum
Propagation delay time (t_{PLH}) A/C and B/C C_L = 50 pF E/C C_L = 50 pF	 2 ms minimum, 5.2 ms maximum 4.8 ms minimum, 120 ms maximum
Propagation delay time (t_{PHL}) A/C and B/C, E/C C_L = 50 pF	 6.3 ms minimum, 27.1 ms maximum
Schmitt trigger hysteresis check (measuring t_{PHL} at V_{IH} = threshold voltage)	1.9 ms minimum, 16.7 ms maximum
Transition time (t_{TLH}) C_L = 50 PF	3.0 ns minimum, 50.0 ns maximum
Transition time (t_{THL}) C_L = 50 pF	3.0 ns minimum, 30.0 ns maximum

1.3 Module function. The interface driver module, designated as key code EKB, is a plug-in Standard Electronic Modules Program (SEMP) electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains eight 28 V/5 V isolator circuits, each consisting of a light emitting diode (LED) optically coupled to an amplifier with a photoreceiver input and TTL compatible output. The input to each LED is current limited by pin programmable resistors. Depending on the resistor chosen, the LED can be operated from 28 Vdc (A/C input), 5 Vdc (B/C input), or 115 Vrms, 400 Hz (E/C input, available on one circuit only). The input of each circuit is isolated from all others and from all outputs. Input E/C is used to limit the current of a 115 Vrms, 400 Hz, signal to the LED. When this signal is present, the TTL logic output level changes state (after an integration time constant) until the signal is removed. The photoreceiver amplifier operates from a TTL logic power and drives 16 mA at V_{OL} = 0.4 V. Hysteresis and a large RC time constant are incorporated into each amplifier to filter input noise from the output (for example, relay or switch bounce).

1.3.1 Input to output relationship. Each isolator circuit has the input to output relationship of table I for positive logic.

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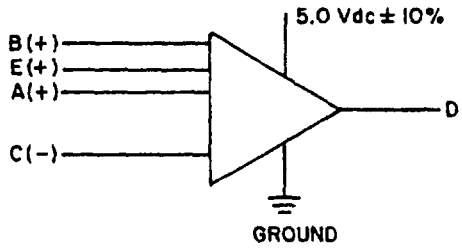


TABLE I. Input to output relationship.

Input	Output
A/C, B/C, E/C	D
V_{IH} or I_{IH}	H
V_{IL} or I_{IL}	L

NOTES:
1. H = High level.
2. L = Low level.

1.3.2 Paralleled outputs. Paralleling outputs is not permissible.

1.3.3 Input polarity. Observe polarity of input voltage.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.759 failures per million operating hours when predicted in accordance with MIL-M-28787.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EKB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

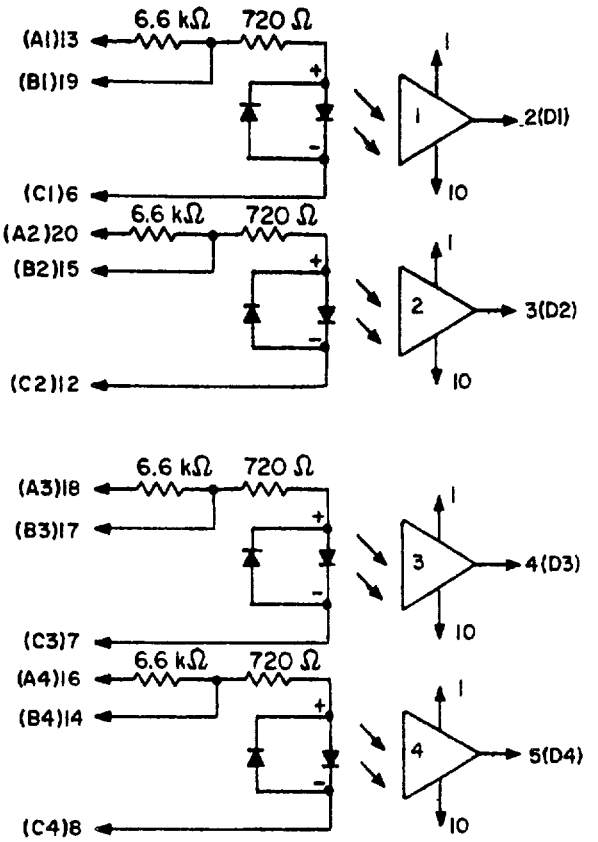
3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/162, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The logic diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

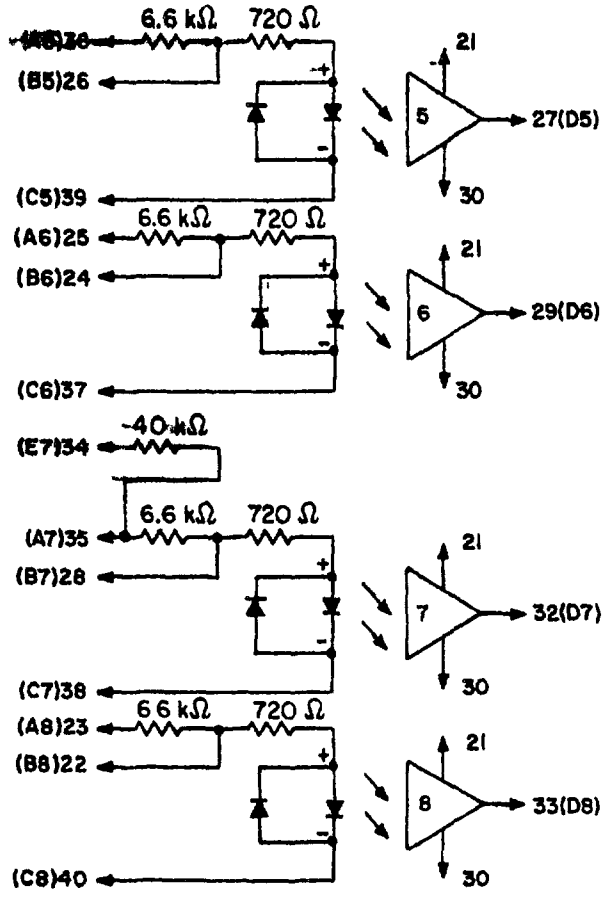
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1 ◄ V_{CC} (CKTS 1, 2, 3, and 4) = 5.0 V ±10%
10 ◄ GND (CKTS 1, 2, 3, and 4) = 0 V REF
11 ◄ MODULE FRAME
9 ◄ UNUSED
31 ◄ UNUSED
21 ◄ V_{CC} (CKTS 5, 6, 7, and 8) = 5.0 V ±10%
30 ◄ GND (CKTS 5, 6, 7, and 8) = 0 V REF

FIGURE 1 EKB module logic diagram, interface driver

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NOTES:
1. A1-A8 = 28 V input.
2. B1-B8 = 5 V input.
3. C1-C8 = A, B, and E signal return.
4. D1-D8 = Circuit output.
5. E7 = 115 Vrms input.
6. Capacitors are 0.01 µF, 50 V.

FIGURE 1. EKB module logic diagram, interface driver - Continued.

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TABLE II. Module electrical requirements

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A11	Isolation (100 V)	III	X	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
A12-A22	Isolation (300 V)	III	S	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
B1-B8	V_{out} (low-loaded)	IV	S	0.02	0.80	0.02	0.80	0.02	0.80	0.02	0.80	V
C1-C8	Output short circuit current	V	X	-18.0	-55.0	-18.0	-55.0	-18.0	-55.0	-18.0	-55.0	mA
D1-D16	Common mode voltage (100 V)	VI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
D17-D32	Common mode voltage (300 V)	VI	S	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
E1-E8	Resistance, 28 V current limiting	VII	X	6.20	7.00	5.94	7.26	6.20	7.00	5.94	7.26	kΩ
E9	Resistance, 115 V current limiting	VII	X	37.6	42.4	36.0	44.0	37.6	42.4	36.0	44.0	kΩ
E10	Continuity, fin to pin 11	VII	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
F1-F8	Input current, high level, A/C input (V_{IH} = 35 V)	VIII	Q	4.27	4.91	4.12	5.13	4.12	5.13	4.12	5.13	mA
G1-G8	Input current, high level, B/C input (V_{IH} = 7.5 V)	IX	Q	7.70	10.00	7.48	10.45	7.48	10.45	7.48	10.45	mA
H	Input current, high level, E/C input (V_{IH} = 126.5 V)	X	Q	2.46	2.85	2.37	3.00	2.37	3.00	2.37	3.00	mA
J1-J8	Input protection diode and 5 V current limiting resistor current	XI	X	2.9	3.8	2.8	3.9	2.7	4.0	2.7	4.0	mA

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
K1-K8	Input voltage, low level, B/C input (I_{IL} = 100 uA)	XII	X	0.7	1.2	0.7	1.2	0.7	1.2	0.7	1.2	V
L1-L8	Output voltage, low level, I_{IL} = 100 uA	XIII	X	0.00	0.40	0.00	0.40	0.00	0.40	0.00	0.40	V
M1-M8	Output voltage, high level, V_{IH} = 20 V	XIV	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
N1, N2	Power supply current, output high level	XV	S	10	44	10	44	10	44	10	44	mA
N3, N4	Power supply current, output low level	XV	S	20	55	20	55	20	55	20	55	mA
P1-P8	Transition time (t_{TLH}), C_L = 50 pF	XVI	S	3	50	3	50	3	50	3	50	ns
Q1-Q8	Transition time (t_{THL}), C_L = 50 pF	XVI	S	3	30	3	30	3	30	3	30	ns
R1-R8	Propagation delay time (t_{PLH}), B/C input, C_L = 50 pF	XVI	X	2.5	4.6	2.0	5.2	2.5	4.6	2.0	5.2	ms
S1-S8	Propagation delay time (t_{PHL}), B/C input, C_L = 50 pF	XVI	X	8.8	20.4	6.6	23.0	8.4	24.0	6.3	27.1	ms
T	Propagation delay time (t_{PLH}), 400 Hz, E/C input, C_L = 50 pF	XVIII	X	4.8	35.0	4.8	35.0	4.8	120.0	4.8	120.0	ms

See notes at end of table.

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TABLE II. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
U1-U8	Schmitt trigger hysteresis check (measuring propagation delay time t_{PHL}, C_L = 50 pF)	XVIII	Q	1.9	14.3	1.6	16.1	1.8	14.9	1.5	16.7	ms
V1-V8	Output crosstalk, low level	XIX	Q	0	400	0	400	0	400	0	400	mV
W1-W8	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV
X	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	mV
Y1, Y2	Power supply filter capacitance	XXI	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	uF
Z	Input short term overload	XXII	S	Must pass all functional tests - see 4.4.2 for order of testing								

NOTES:

1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.

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3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II.

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality Assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below.

Equipment	MIL-STD-1665	No. required
DC power supply	Item P01	3
Pulse generator	Item J04	2
DC power supply	Item P03	1
Megohm bridge	Item R06	1
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Ohmmeter	Item R01	1
Oscilliscope	Item D01	1
Oscilliscope probe	Item Y02	2
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	1
Counter-timer	Item T01	1
Oscilloscope probe	Item Y03	1
Capacitance bridge	Item C01	1
DC power supply	Item P04	1
Pulse generator	Item J02	1
Oscilloscope	Item D03	1

Resistors	No. required
316Ω ±1%, 1/4 W	8
3.01 kΩ ±1%, 1/4 W	8
Value as required for figure 10, note 3, 1/4 W	8
200 kΩ ±1%, 1/8 W	1
51Ω ±5%, 2 W	1

Capacitor	
Value as required for figure 10, note 4	8

Diode	
1N4150, or equivalent	8

Transistor	
2N1893, or equivalent	1

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions: (1) If test condition Z is part of a test sequence, it shall be performed first. (2) If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through Y shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column, and test conditions Y1 and Y2, shall be performed using the initial limits at 25°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test, electrical tests A, and C through Y, shall be performed in the order specified in 4.4.2 using the 25° ±5°C end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life.

6.3 0 volt reference. The reference for test setups will be 0 volt and not necessarily earth ground.

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6.4 Decoupling. Decoupling of test voltages and V_{CC} to minimize noise in the test setups is permissible except for test condition X, power supply transient amplitude test.

Preparing activity:
Navy - EC

Agent:
DLA - ES

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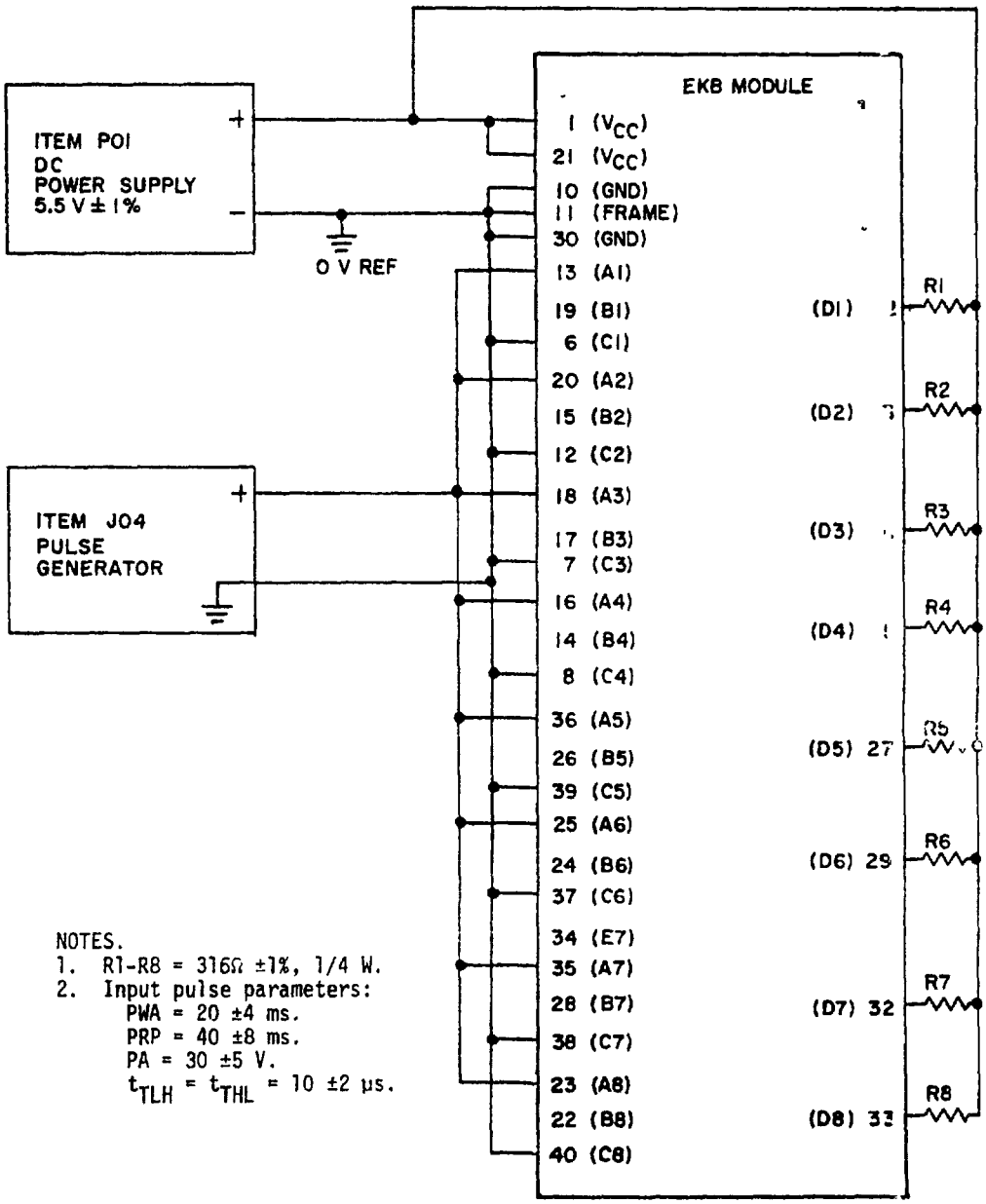
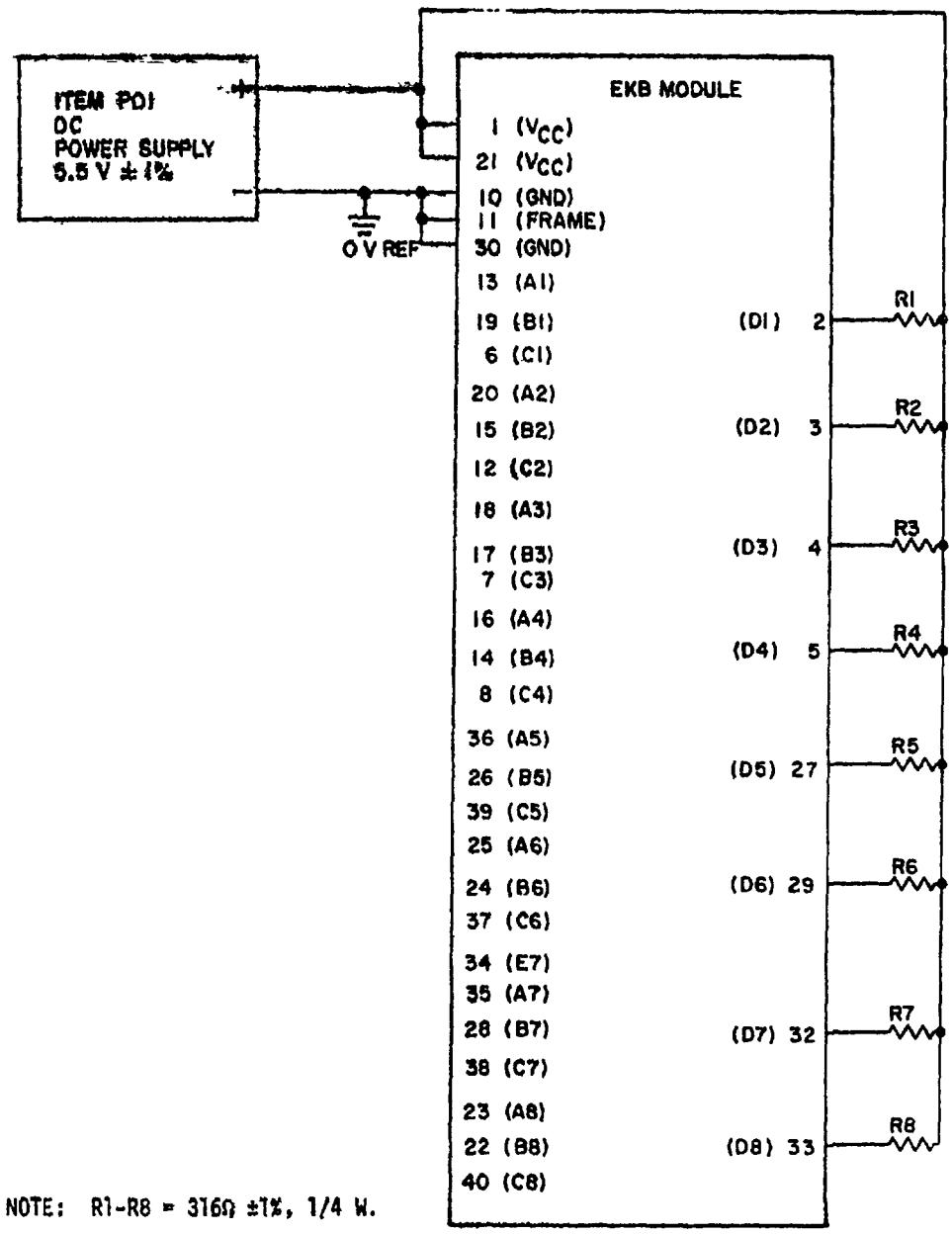


FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R8 = 316Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit A.

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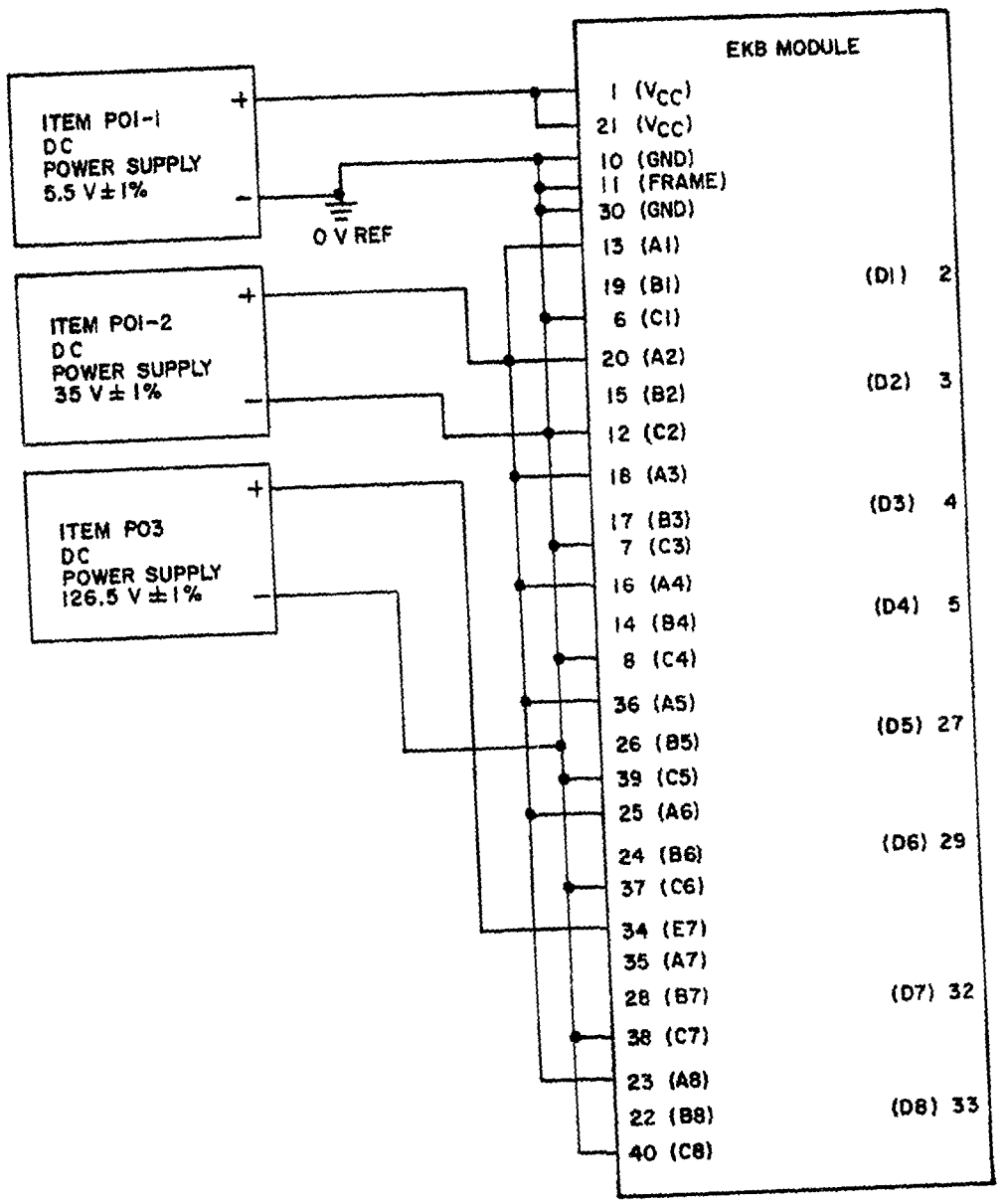


FIGURE 4. Component temperature test circuit B.

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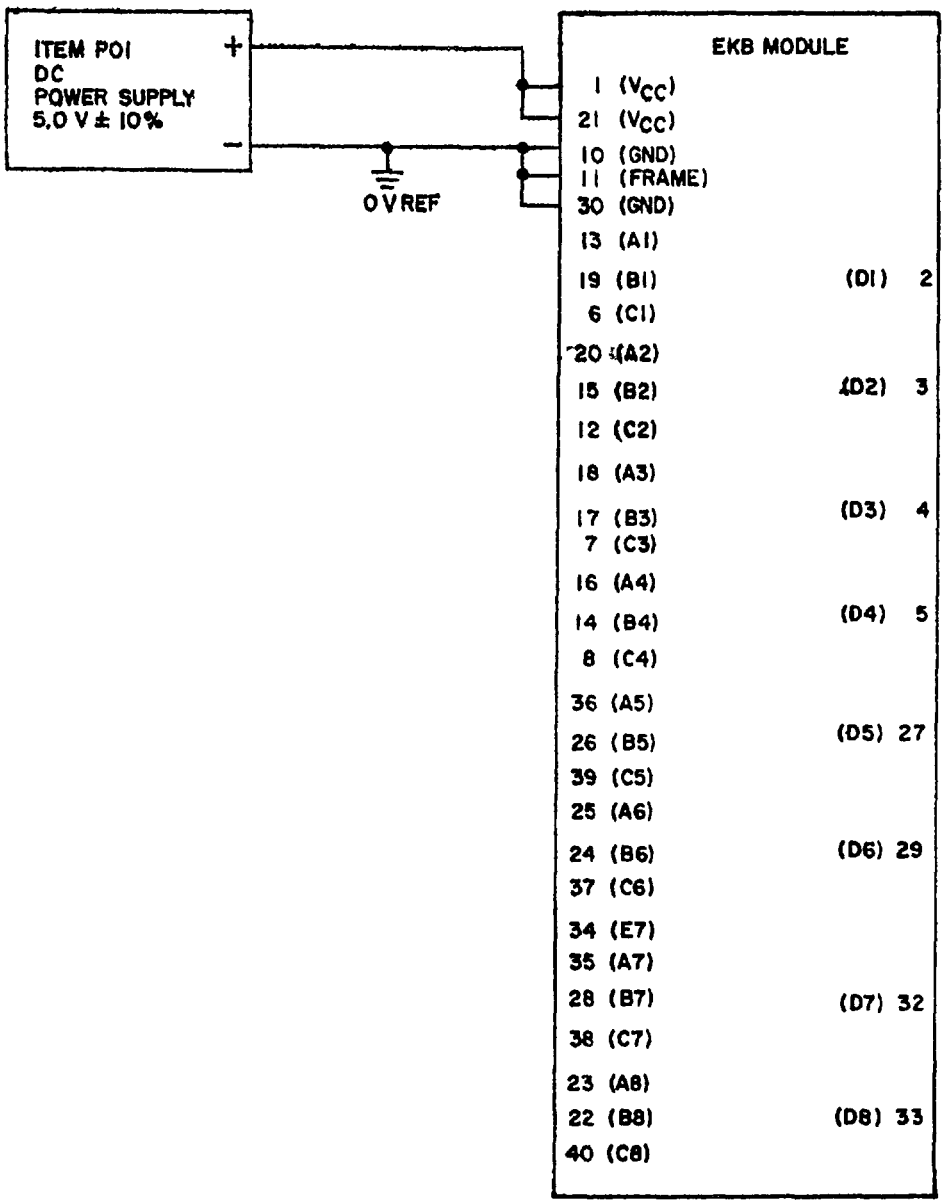


FIGURE 5. Vibration test circuit.

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TABLE III. Isolation test

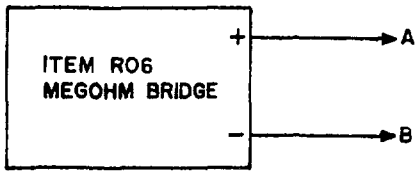
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation (100 V), all pins to pin 11	All pins except pin 11	11	10.0	∞	MΩ	6
A2	Isolation (100 V), unused pins	9, 31	All other module pins				
A3	Isolation (100 V), V_{CC} input and unused pins	1, 10, 9	21, 30, 31				
A4	Circuit 1 Isolation (100 V), input network	6, 13, 19	All other module pins				
A5	Circuit 2	12, 15, 20					
A6	Circuit 3	7, 17, 18					
A7	Circuit 4	8, 14, 16					
A8	Circuit 5	26, 36, 39					
A9	Circuit 6	24, 25, 37					
A10	Circuit 7	28, 34, 35, 38					
A11	Circuit 8 Isolation (100 V), input network	22, 23, 40	All other module pins				
A12	Isolation (300 V), all pins to pin 11	All pins except pin 11	11	10.0	∞	MΩ	6

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TABLE III. Isolation test - Continued

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A13	Isolation (300 V), unused pins	9, 31	All other module pins	10.0	∞	MΩ	6
A14	Isolation (300 V), V_{CC} input and unused pins	1, 10, 9	21, 30, 31				
A15	Circuit 1 Isolation (300 V), input network	6, 13, 19	All other module pins				
A16	Circuit 2	12, 15, 20					
A17	Circuit 3	7, 17, 18					
A18	Circuit 4	8, 14, 16					
A19	Circuit 5	26, 36, 39					
A20	Circuit 6	24, 25, 37					
A21	Circuit 7	28, 34, 35 38					
A22	Circuit 8 Isolation (300 V), input network	22, 23, 40	All other module pins	10.0	∞	MΩ	6

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTES:
1. For test conditions A1-A11, the output voltage of item R06 shall be 100 Vdc ±10%.
2. For test conditions A12-A22, the output voltage of item R06 shall be 300 Vdc ±10%.

EKB MODULE		
1 (V_{CC})		
21 (V_{CC})		
10 (GND)		
11 (FRAME)		
30 (GND)		
13 (A1)		
19 (B1)	(D1)	2
6 (C1)		
20 (A2)		
15 (B2)	(D2)	3
12 (C2)		
18 (A3)		
17 (B3)	(D3)	4
7 (C3)		
16 (A4)		
14 (B4)	(D4)	5
8 (C4)		
36 (A5)		
26 (B5)	(D5)	27
39 (C5)		
25 (A6)		
24 (B6)	(D6)	29
37 (C6)		
34 (E7)		
35 (A7)		
28 (B7)	(D7)	32
38 (C7)		
23 (A8)		
22 (B8)	(D8)	33
40 (C8)		

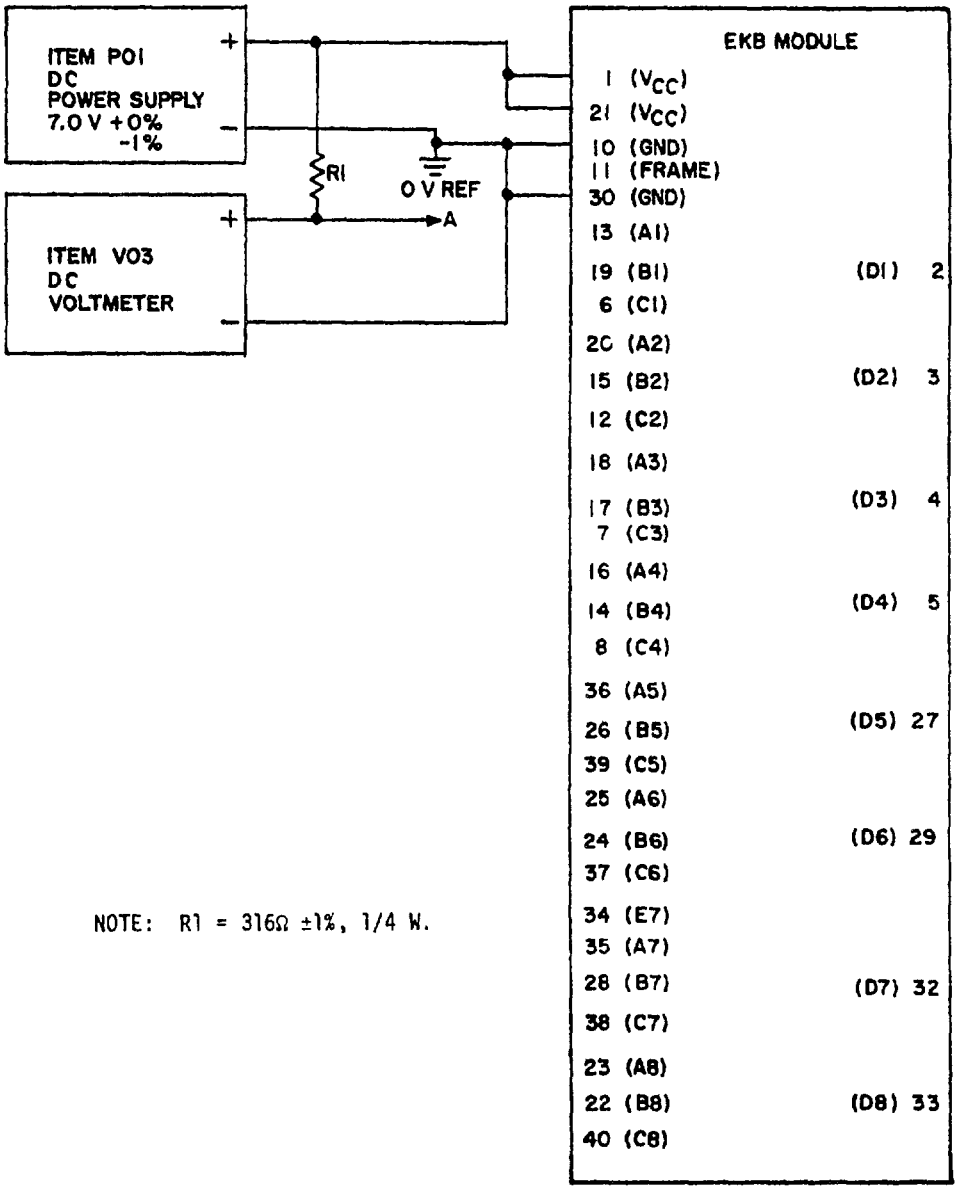
FIGURE 6 Isolation test circuit for table III.

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TABLE IV. V_{out} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B1	D1 V_{out} (low-loaded)	2	0.02	0.80	V	7
B2	D2	3				
B3	D3	4				
B4	D4	5				
B5	D5	27				
B6	D6	29				
B7	D7	32				
B8	D8 V_{out} (low-loaded)	33	0.02	0.80	V	7

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NOTE: R1 = 316Ω ±1%, 1/4 W.

FIGURE 7. V_{out} (low loaded) test circuit for table IV.

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TABLE V. Output short circuit current test

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
C1	D1 Output short circuit current	2	-18.0	-55.0	mA	8
C2	D2	3				
C3	D3	4				
C4	D4	5				
C5	D5	27				
C6	D6	29				
C7	D7	32				
C8	D8 Output short circuit current	33	-18.0	-55.0	mA	8

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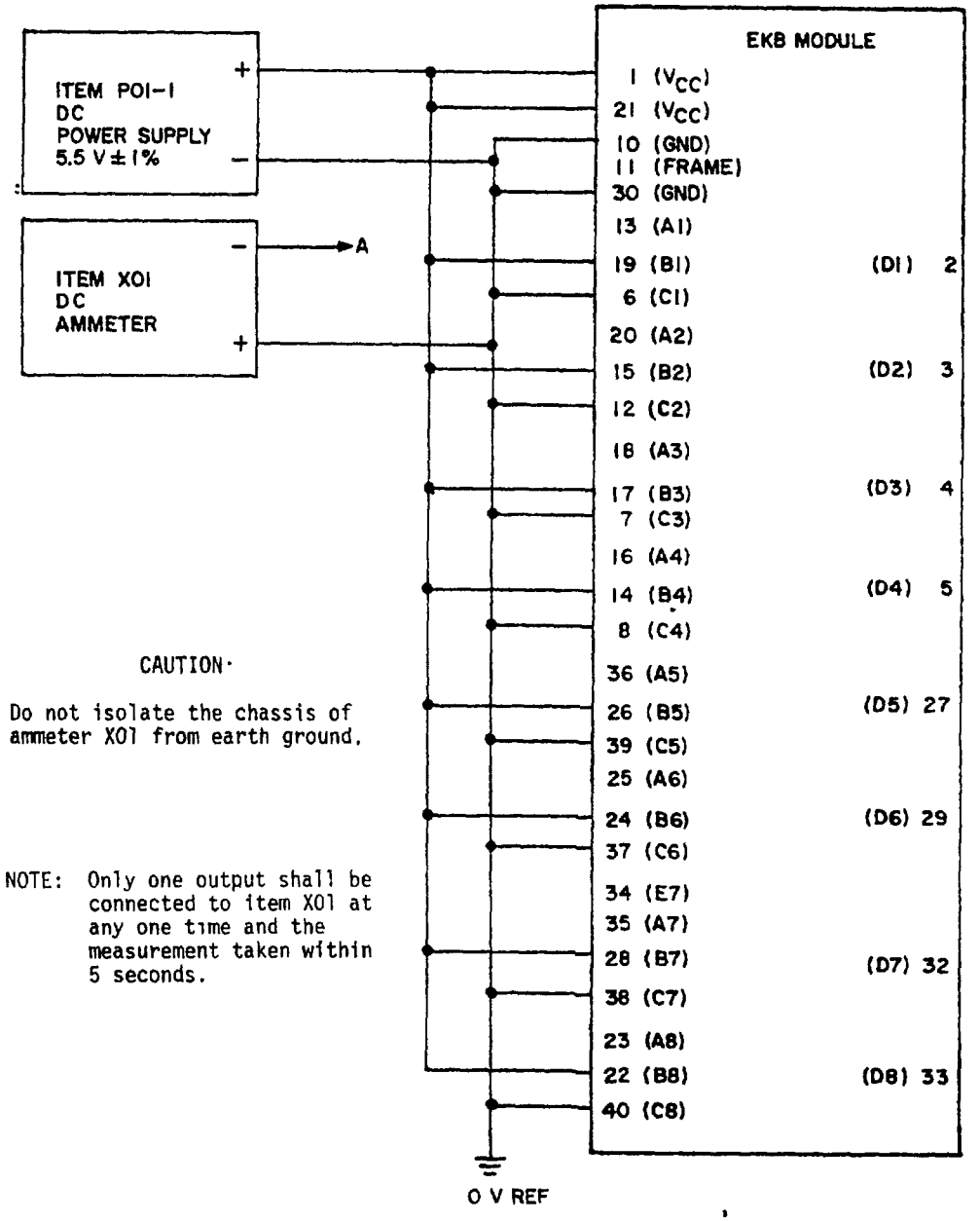


FIGURE 8. Output short circuit current test circuit for table V.

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TABLE VI. Common mode voltage test.

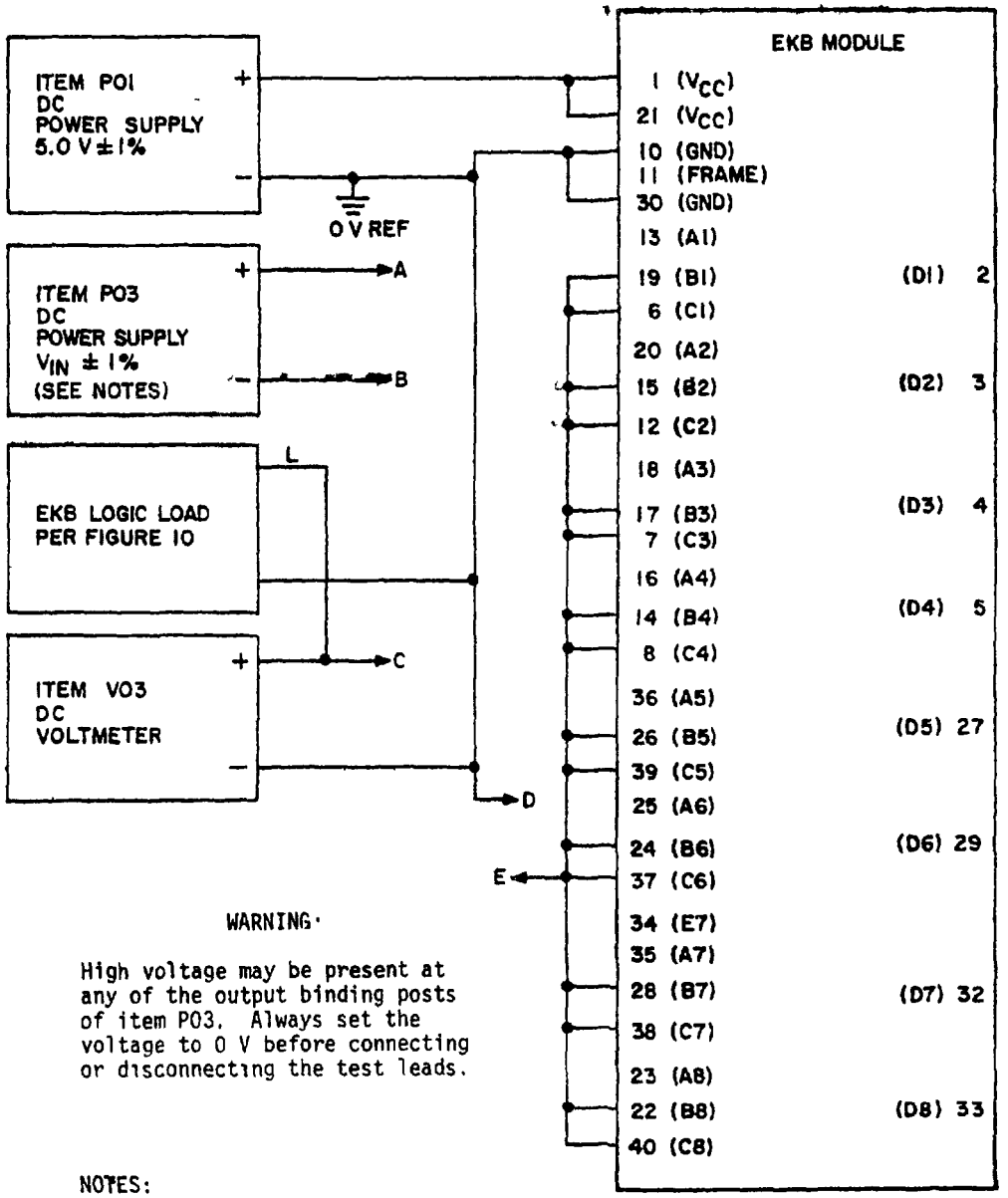
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	D1 Common mode voltage (100 V)	E	D	2	0.0	0.4	V	9
D2	D2			3				
D3	D3			4				
D4	D4			5				
D5	D5			27				
D6	D6			29				
D7	D7			32				
D8	D8	E	D	33				
D9	D1	D	E	2				
D10	D2			3				
D11	D3			4				
D12	D4			5				
D13	D5			27				
D14	D6			29				
D15	D7			32				
D16	D8 Common mode voltage (100 V)	D	E	33				
D17	D1 Common mode voltage (300 V)	E	D	2	0.0	0.4	V	9

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TABLE VI. Common mode voltage test - Continued

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D18	D2 Common mode voltage (300 V)	E	D	3	0.0	0.4	V	9
D19	D3			4				
D20	D4			5				
D21	D5			27				
D22	D6			29				
D23	D7			32				
D24	D8	E	D	33				
D25	D1	D	E	2				
D26	D2			3				
D27	D3			4				
D28	D4			5				
D29	D5			27				
D30	D6			29				
D31	D7			32				
D32	D8 Common mode voltage (300 V)	D	E	33	0.0	0.4	V	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:22Z
Check the source to verify that this is the current version before use.



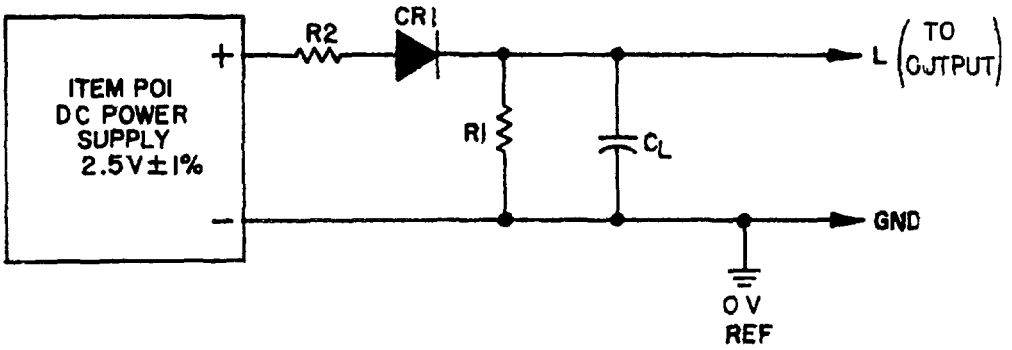
WARNING:

High voltage may be present at any of the output binding posts of item P03. Always set the voltage to 0 V before connecting or disconnecting the test leads.

NOTES:
1. For test conditions D1-D16, V_{IN} = 100 V ±1%.
2. For test conditions D17-D32, V_{IN} = 300 V ±1%.

FIGURE 9. Common mode voltage test circuit for table VI.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. R1 = 3.01 kΩ ±1%, 1/4 W.
3. R2 = Select value for a current flow out of point L of 16.0 mA ±1% with po nt L held at 0.4 V ±1%.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance). C_L is not requ red for test conditions D, L, and M.
5. Lead length from load circuit to module output pin under test shall be les' than 1 inch (25.4 mm).

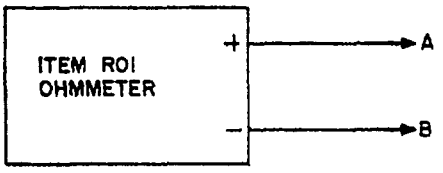
FIGURE 10. EKB logic load circuit.

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TABLE VII. Resistance and continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	A/B 1 Resistance, 28 V current limiting	13	19	6.20	7.00	kΩ	11
E2	A/B 2	20	15				
E3	A/B 3	18	17				
E4	A/B 4	16	14				
E5	A/B 5	36	26				
E6	A/B 6	25	24				
E7	A/B 7	35	28				
E8	A/B 8 Resistance, 28 V current limiting	23	22	6.20	7.00		
E9	E/A 7 Resistance, 115 V current limiting	34	35	37.6	42.4	kΩ	
E10	Continuity, fin to pin 11	11	Fin	0.0	1.0	Ω	11

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NOTE: For test condition E10, penetrate the anodized finish on the module fin with a sharply pointed probe connected to the negative terminal.

EKB MODULE		
1 (V_{CC})		
21 (V_{CC})		
10 (GND)		
11 (FRAME)		
30 (GND)		
13 (A1)		
19 (B1)	(D1)	32
6 (C1)		
20 (A2)		
15 (B2)	(D2)	3
12 (C2)		
18 (A3)		
17 (B3)	(D3)	4
7 (C3)		
16 (A4)		
14 (B4)	(D4)	5
8 (C4)		
36 (A5)		
26 (B5)	(D5)	27
39 (C5)		
25 (A6)		
24 (B6)	(D6)	29
37 (C6)		
34 (E7)		
35 (A7)		
28 (B7)	(D7)	32
38 (C7)		
23 (A8)		
22 (B8)	(D8)	33
40 (C8)		

FIGURE 11. Resistance and continuity test circuit for table VII

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TABLE VIII. Input current, high level, A/C input (V_{IH} = 35 V), test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
F1	A/C 1 Input current, high level, V_{IH} = 35 V	13	4.27	4.91	mA	12
F2	A/C 2	20				
F3	A/C 3	18				
F4	A/C 4	16				
F5	A/C 5	36				
F6	A/C 6	25				
F7	A/C 7	35				
F8	A/C 8 Input current, high level, V_{IH} = 35 V	23	4.27	4.91	mA	12

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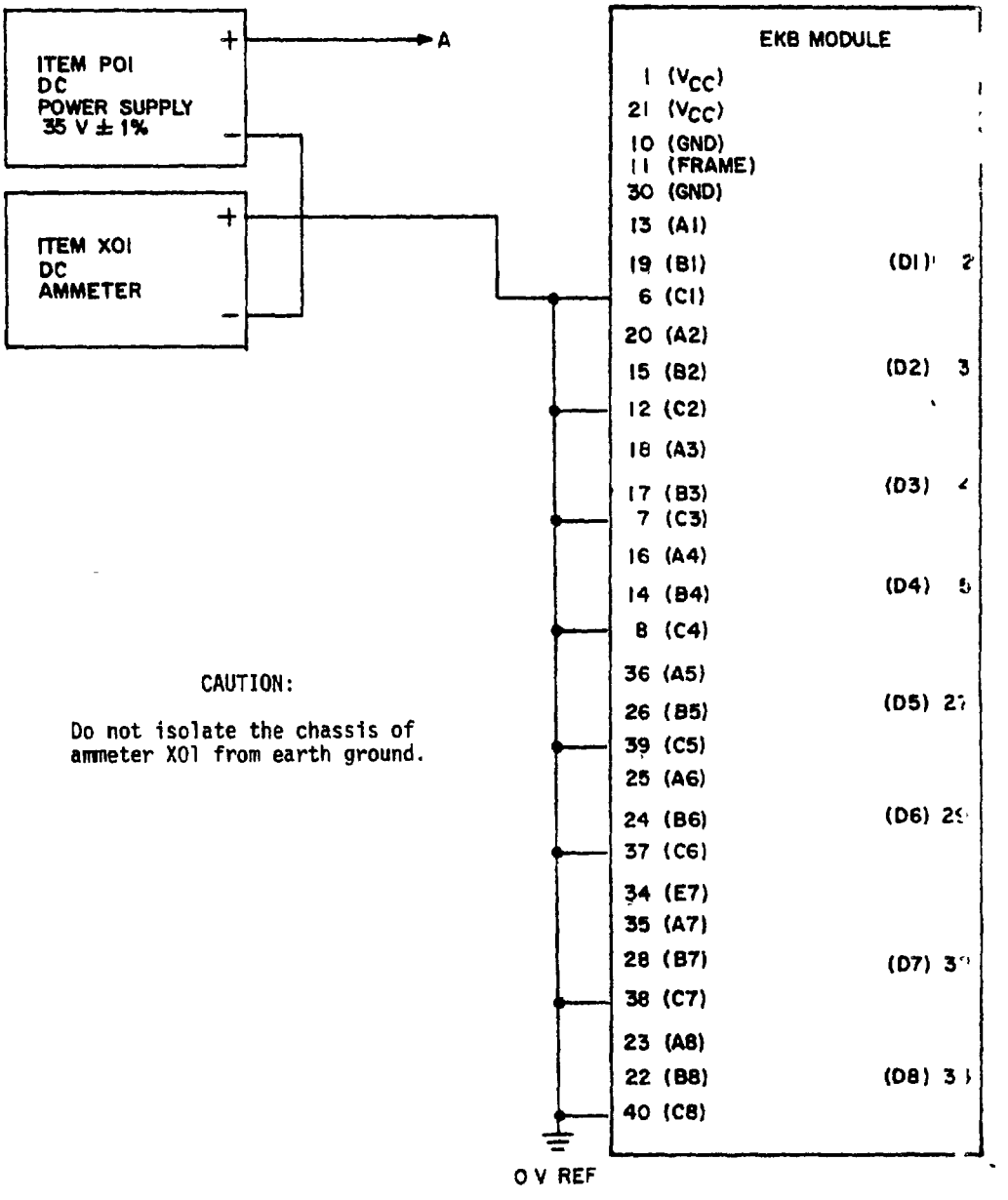


FIGURE 12 Input current, high level, A/C input (V_{IH} = 35 V), test circuit for table VIII.

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TABLE IX. Input current, high level, B/C input (V_{IH} = 7.5 V), test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
G1	B/C 1 Input current, high level, V_{IH} = 7.5 V	19	7.70	10.00	mA	13
G2	B/C 2	15				
G3	B/C 3	17				
G4	B/C 4	14				
G5	B/C 5	26				
G6	B/C 6	24				
G7	B/C 7	28				
G8	B/C 8 Input current, high level, V_{IH} = 7.5 V	22	7.70	10.00	mA	13

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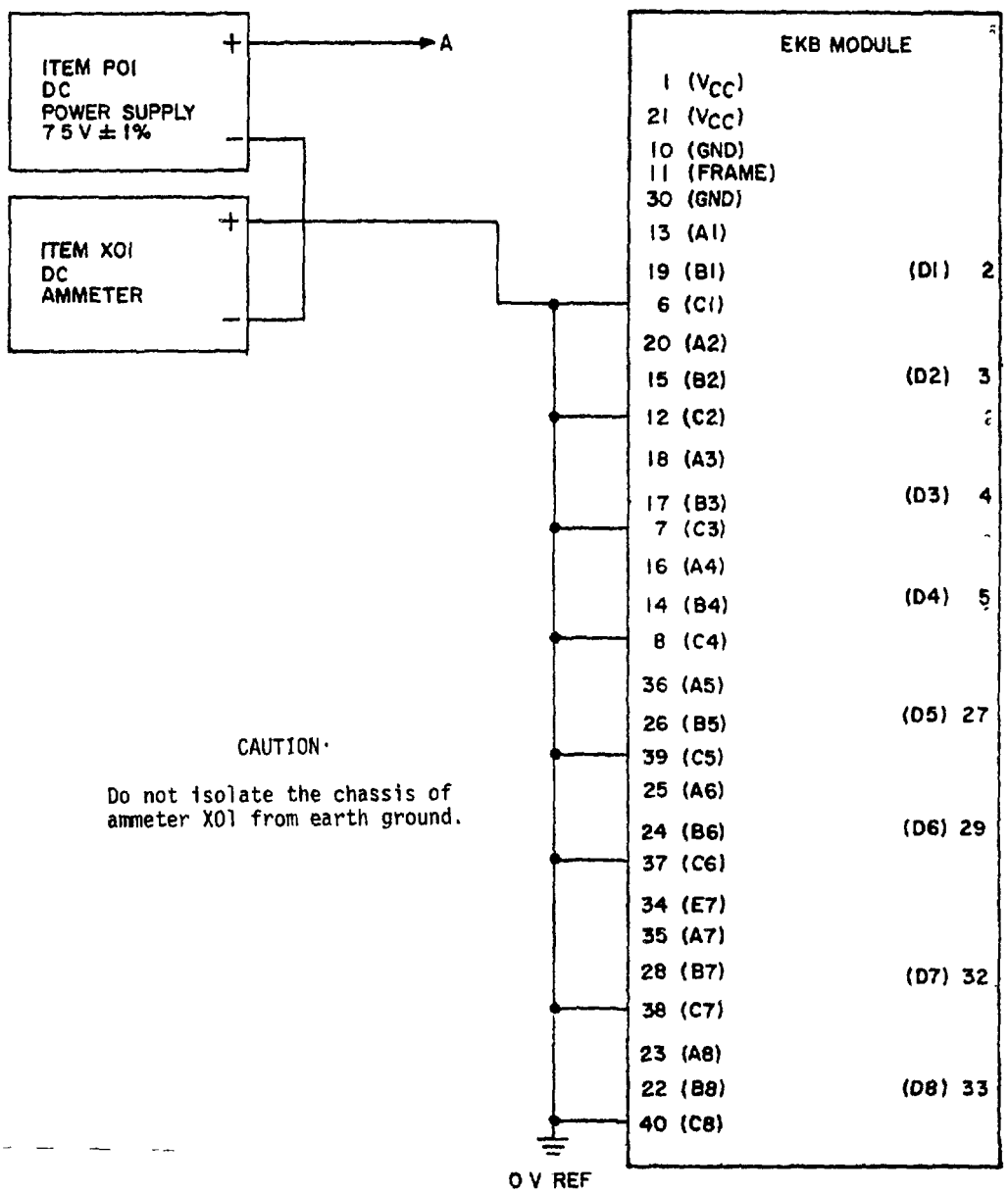


FIGURE 13. Input current, high level, B/C input (V_{IH} = 7.5 V) test circuit for table IX.

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TABLE X. Input current, high level, E/C input (V_{IH} = 126.5 V), test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
H	E/C 7 Input current, high level, V_{IH} = 126.5 V	2.46	2.85	mA	14

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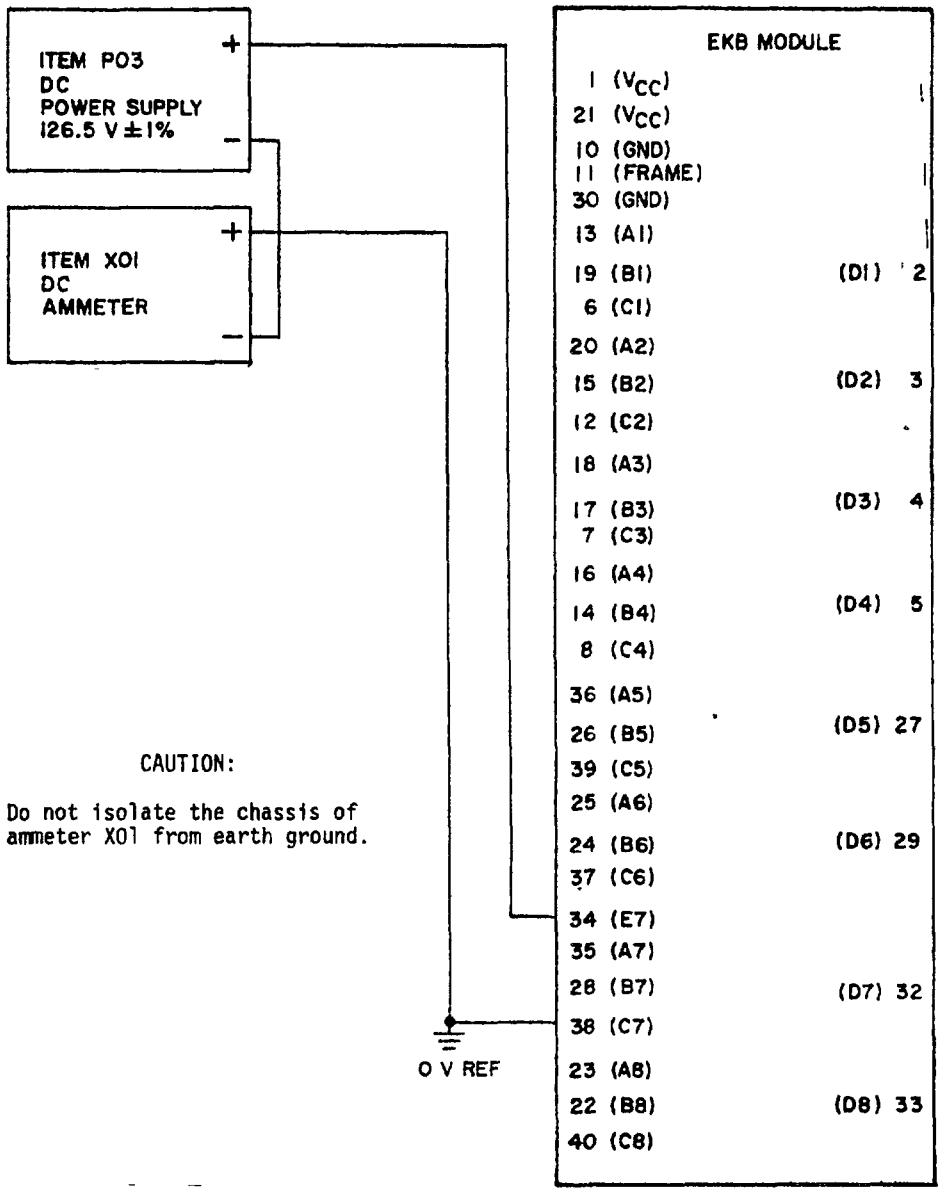


FIGURE 14. Input current, high level, E/C input (V_{IH} = 126.5 V) test circuit for table X.

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TABLE XI. Input protection diode and 5 V current limiting resistor current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
J1	B/C 1 Input protection diode and 5 V current limiting resistor current	19	2.9	3.8	mA	15
J2	B/C 2	15				
J3	B/C 3	17				
J4	B/C 4	14				
J5	B/C 5	26				
J6	B/C 6	24				
J7	B/C 7	28				
J8	B/C 8 Input protection diode and 5 V current limiting resistor current	22	2.9	3.8	mA	15

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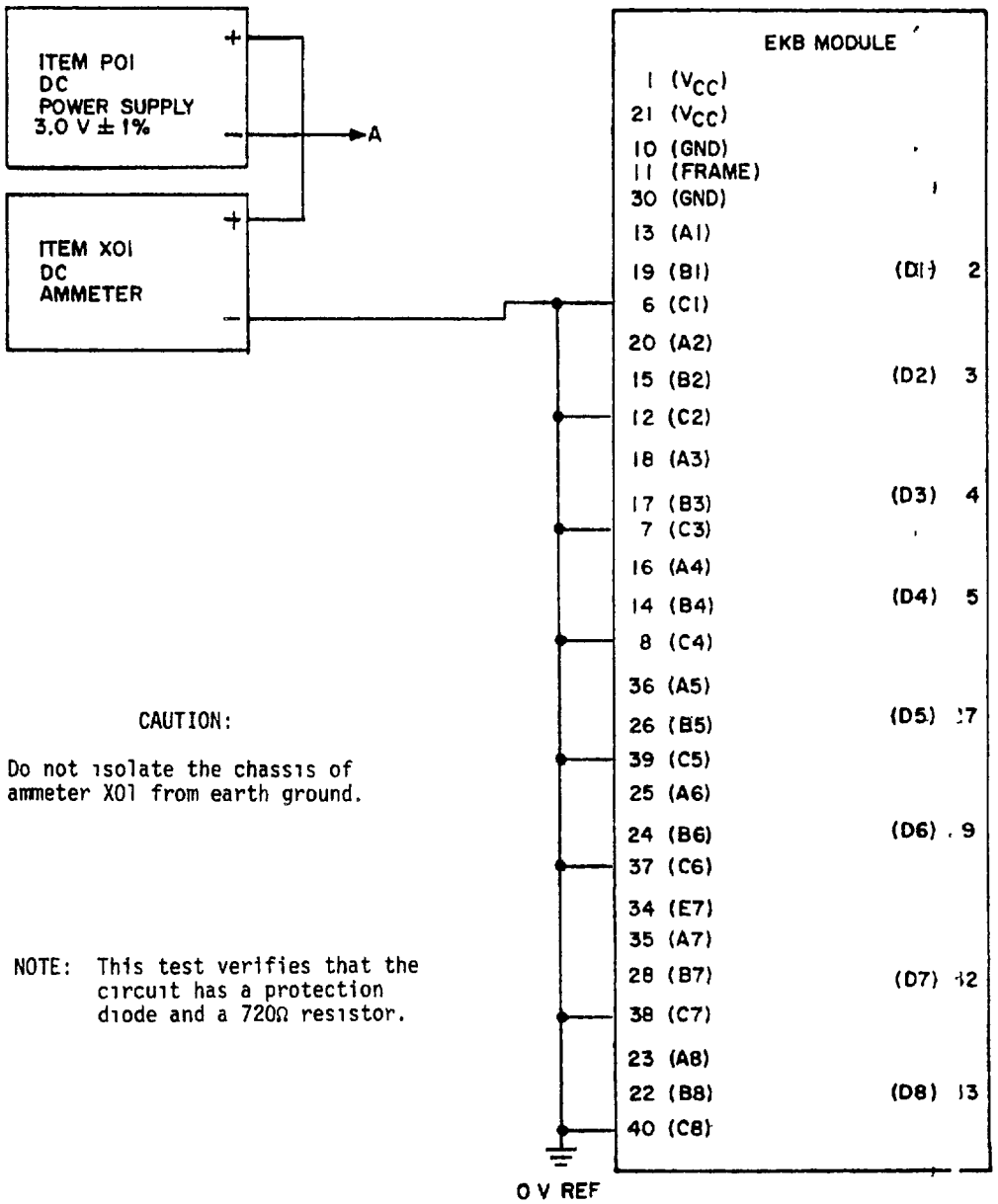


FIGURE 15. Input protection diode and 5 V current limiting resistor current test circuit for table XI.

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TABLE XII. Input voltage, low level, B/C input (I_{IL} = 100 uA), test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	B/C 1 Input voltage, low level, I_{IL} = 100 uA	19	Open	19	Open	0.7	1.2	V	16
K2	B/C 2	15		15					
K3	B/C 3	17		17					
K4	B/C 4	14		14					
K5	B/C 5	26		26					
K6	B/C 6	24		24					
K7	B/C 7	28		28					
K8	B/C 8 Input voltage, low level, I_{IL} = 100 uA	22	Open	22	Open	0.7	1.2	V	16

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TABLE XIII. Output voltage, low level, I_{IL} = 100 uA, test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	D1 Output voltage, low level, I_{IL} = 100 uA	19	20, 18, 16, 36, 25, 35, 23	2	2	0.00	0.40	V	16
L2	D2	15	13, 18, 16, 36, 25, 35, 23	3	3				
L3	D3	17	13, 20, 16, 36, 25, 35, 23	4	4				
L4	D4	14	13, 20 18, 36, 25, 35, 23	5	5				
L5	D5	26	13, 20, 18, 16, 25, 35, 23	27	27				
L6	D6	24	13, 20, 18, 16, 36, 35, 23	29	29				
L7	D7	28	13, 20, 18, 16, 36, 25, 23	32	32				
L8	D8 Output voltage, low level, I_{IL} = 100 uA	22	13, 20, 18, 16, 36, 25, 35	33	33	0.00	0.40	V	16

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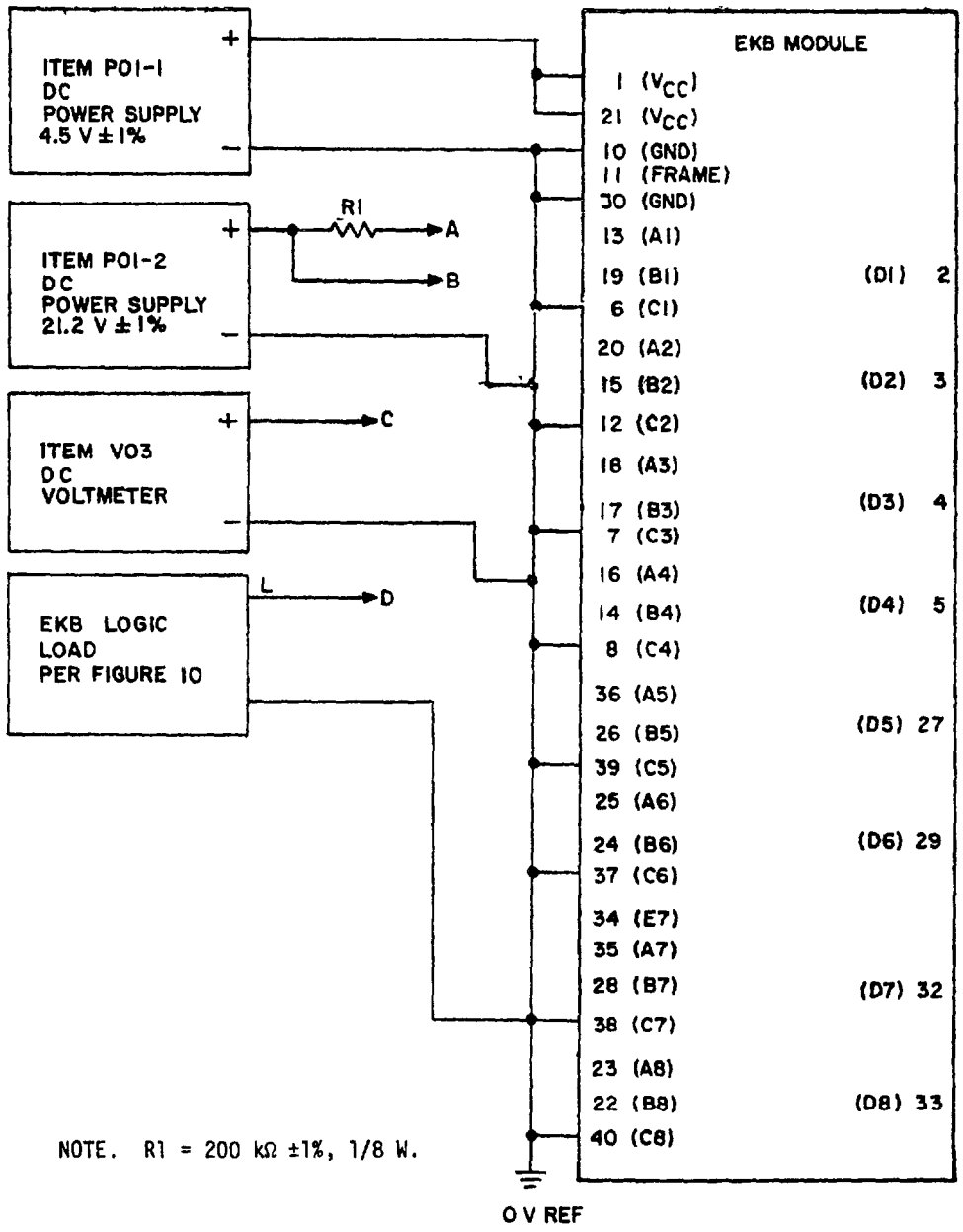


FIGURE 16. Input voltage, low level, B/C input (I_{IL} = 100 µA) and output voltage, low level, I_{IL} = 100 µA, test circuit for tables XII and XIII.

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TABLE XIV. Output voltage, high level, V_{IH} = 20 V, test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	D1 Output voltage, high level, V_{IH} = 20 V	13	2	2.4	4.5	V	17
M2	D2	20	3				
M3	D3	18	4				
M4	D4	16	5				
M5	D5	36	27				
M6	D6	25	29				
M7	D7	35	32				
M8	D8 Output voltage, high level, V_{IH} = 20 V	23	33	2.4	4.5	V	17

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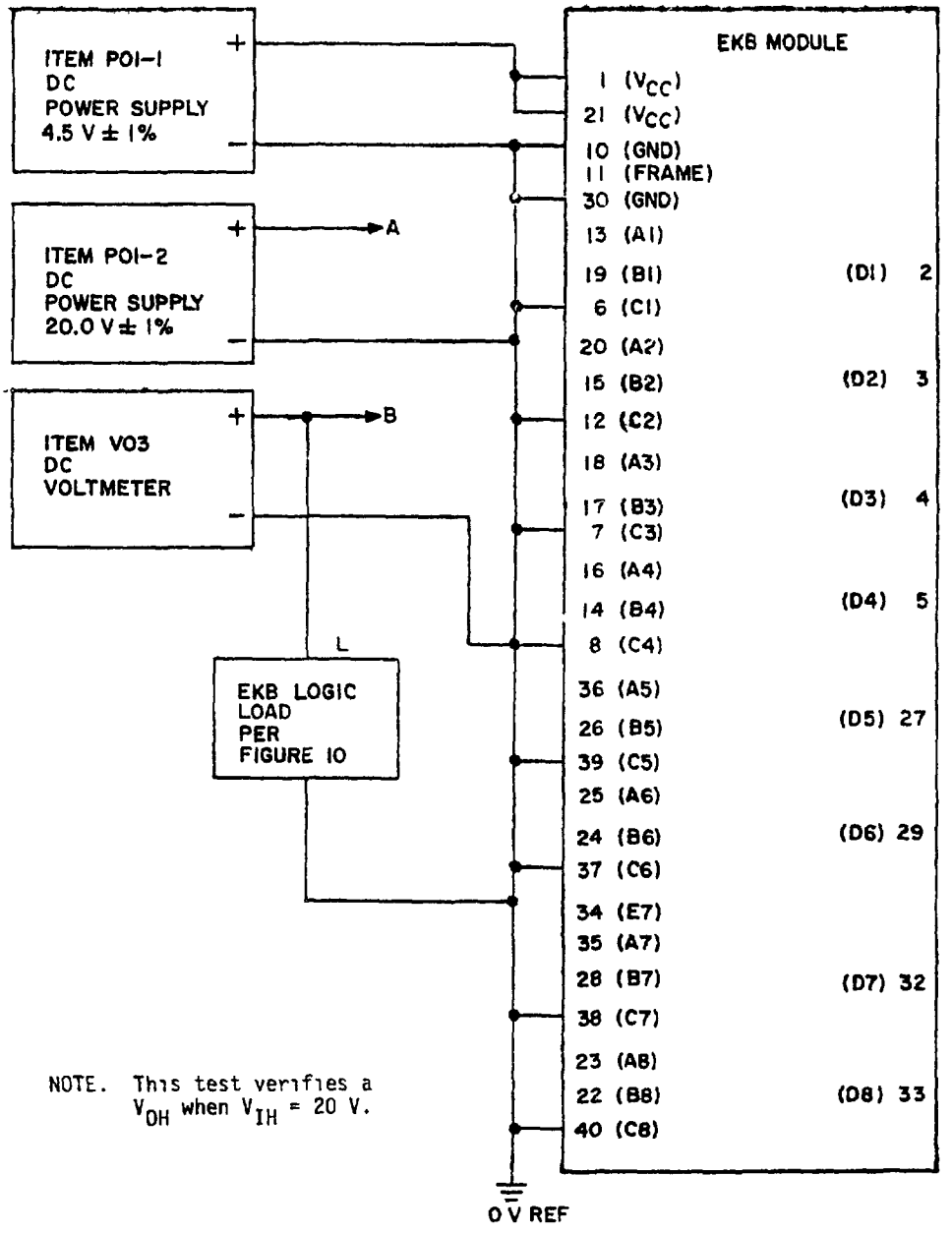


FIGURE 17. Output voltage, high level, V_{IH} = 20 V, test circuit for table XIV.

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TABLE XV. Power supply current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Power supply current, output high level	1	C	10	44	mA	18
N2	Power supply current, output high level	21	C	10	44		
N3	Power supply current, output low level	1	Open	20	55		
N4	Power supply current, output low level	21	Open	20	55	mA	18

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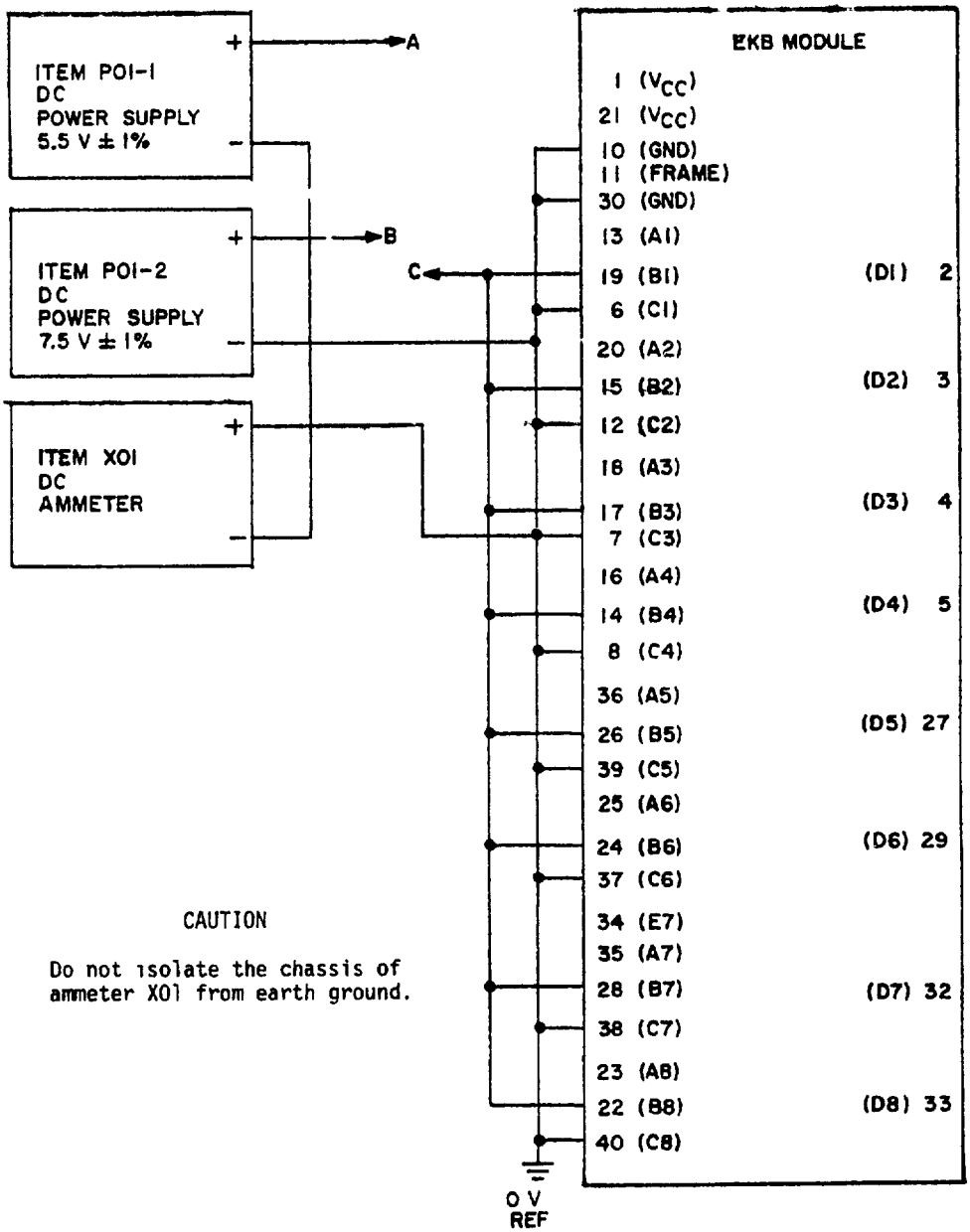


FIGURE 18. Power supply current, output high-level and output low-level, test circuit for table XV.

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TABLE XVI. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

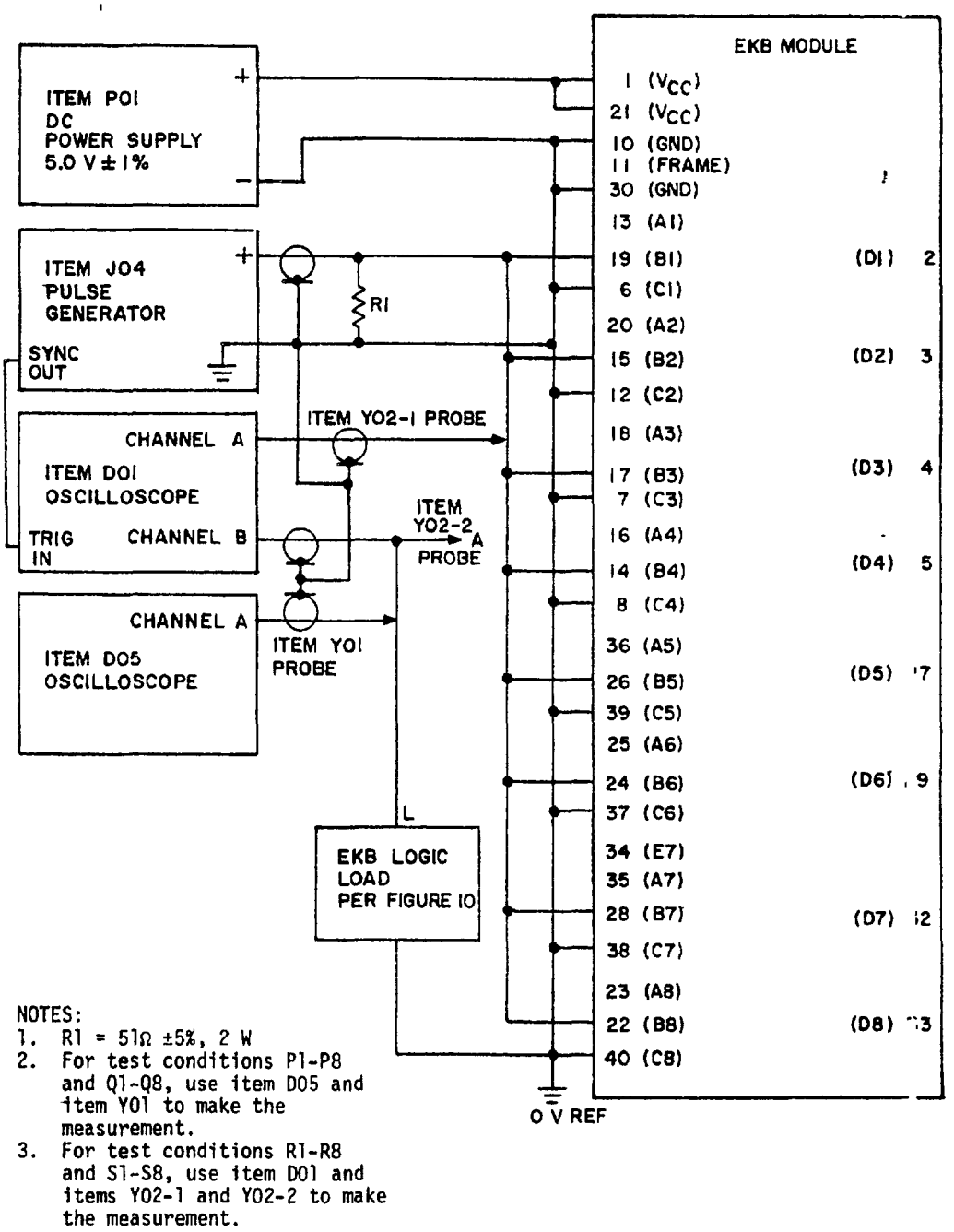
Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
P1	D1 Transition time (t_{TLH}), C_L = 50 pF	2	3	50	ns	19,20
P2	D2	3				
P3	D3	4				
P4	D4	5				
P5	D5	27				
P6	D6	29				
P7	D7	32				
P8	D8 Transition time (t_{TLH}), C_L = 50 pF	33		50		
Q1	D1 Transition time (t_{THL}), C_L = 50 pF	2		30		
Q2	D2	3				
Q3	D3	4				
Q4	D4	5				
Q5	D5	27				
Q6	D6	29				
Q7	D7	32				
Q8	D8 Transition time (t_{THL}), C_L = 50 pF	33	3	30	ns	19,20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
R1	Propagation delay time (t_{PLH}), B/C input, C_L = 50 pF	2	2.5	4.6	ms	19,20
R2	↓	3	↓	↓	↓	↓
R3		4				
R4		5				
R5		27				
R6		29				
R7		32				
R8	Propagation delay time (t_{PLH}), B/C input, C_L = 50 pF	33	2.5	4.6		
S1	Propagation delay time (t_{PHL}), B/C input, C_L = 50 pF	2	8.8	20.4		
S2	↓	3	↓	↓		
S3		4				
S4		5				
S5		27				
S6		29				
S7		32				
S8	Propagation delay time (t_{PHL}), B/C input, C_L = 50 pF	33	8.8	20.4	ms	19,20

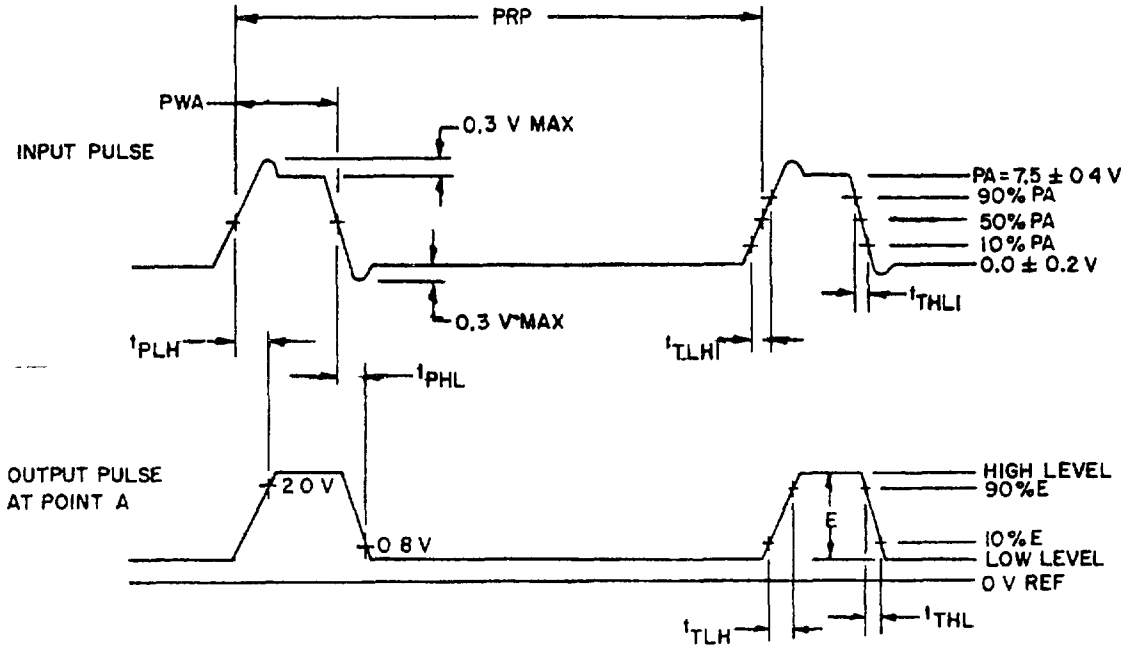
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NOTES:
1. R1 = 51Ω ±5%, 2 W
2. For test conditions P1-P8 and Q1-Q8, use item D05 and item Y01 to make the measurement.
3. For test conditions R1-R8 and S1-S8, use item D01 and items Y02-1 and Y02-2 to make the measurement.

FIGURE 19. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVI

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NOTES:
1. Input pulse parameters for test conditions R1-R8 and S1-S8:
   PWA = 20 ±1 ms.
   PRP = 100 ±5 ms.
   $t_{TLH1} = t_{THL1}$ = 10 ±2 µs.
2. Input pulse parameters for test conditions P1-P8 and Q1-Q8:
   PWA = 6.0 ±0.5 ms
   PRP = 30.0 ±1.0 ms
   $t_{TLH1} = t_{THL1}$ = 10 ±2 µs
3. Verify that all outputs have a high level ≥ 2.4 V and a low level ≤ 0.4 V before making measurements
4. All voltages are referenced to 0 V unless otherwise specified.

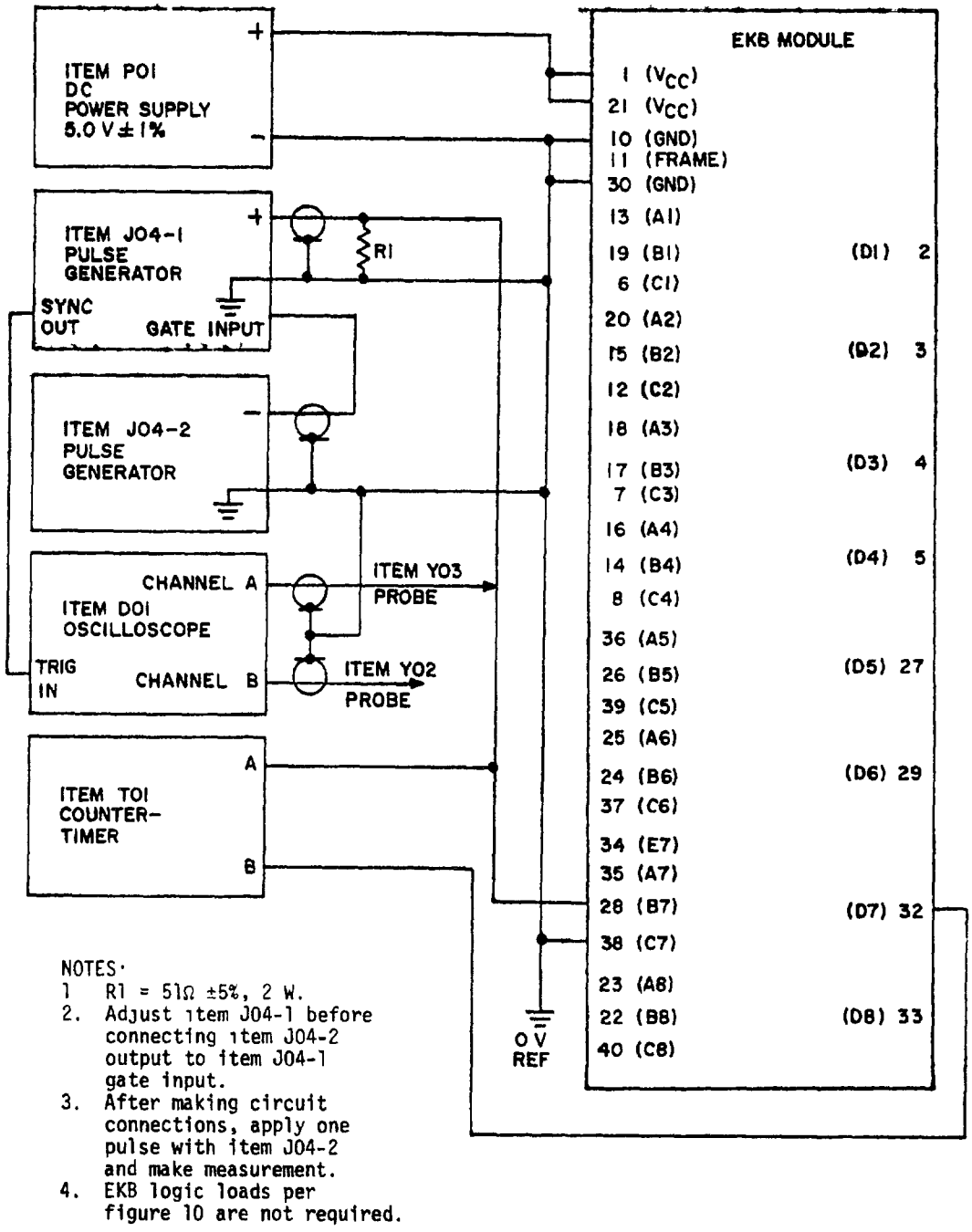
FIGURE 20. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

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TABLE XVII. Propagation delay time (t_{PLH}), E/C input, 400 Hz, test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
T	Propagation delay time (t_{PLH}), E/C input, 400 Hz, C_L = 50 pF	4.8	35.0	ms	21,22

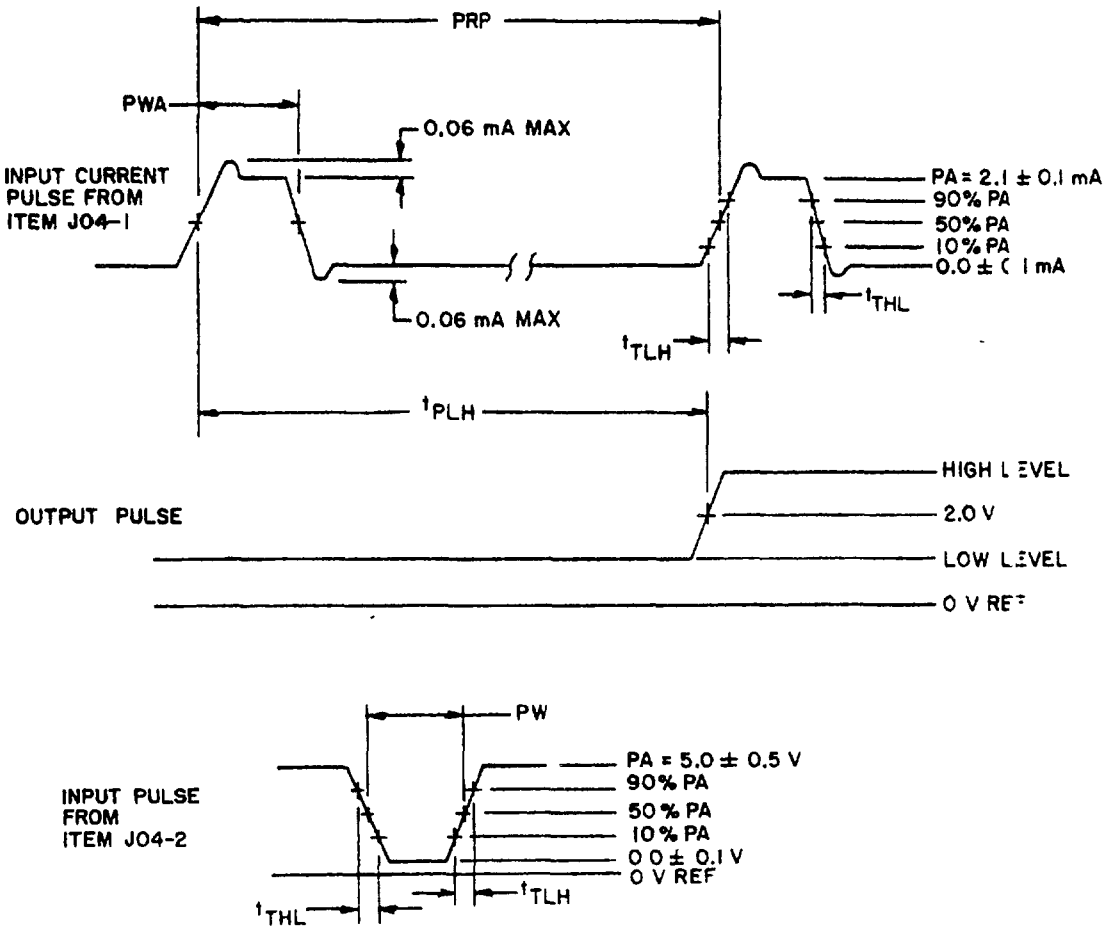
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NOTES:
1. R1 = 51Ω ±5%, 2 W.
2. Adjust item J04-1 before connecting item J04-2 output to item J04-1 gate input.
3. After making circuit connections, apply one pulse with item J04-2 and make measurement.
4. EKB logic loads per figure 10 are not required.

FIGURE 21. Propagation delay time (t_{PLH}), E/C input, 400 Hz, test circuit for table XVII.

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NOTES:
1. Input pulse parameters for item J04-1:
   PWA = 0.95 ±0.05 ms.
   PRP = 2.5 ±0.1 ms.
   $t_{TLH} = t_{THL} = 10 \pm 2$ µs.
2. Input pulse parameters for item J04-2:
   PRP = 240 ±40 ms.
   PW = 120 ±20 ms.
   $t_{THL} = t_{TLH} = 100 \pm 10$ µs.
3. After setting up the pulse, place item J04-2 in the single pulse mode.

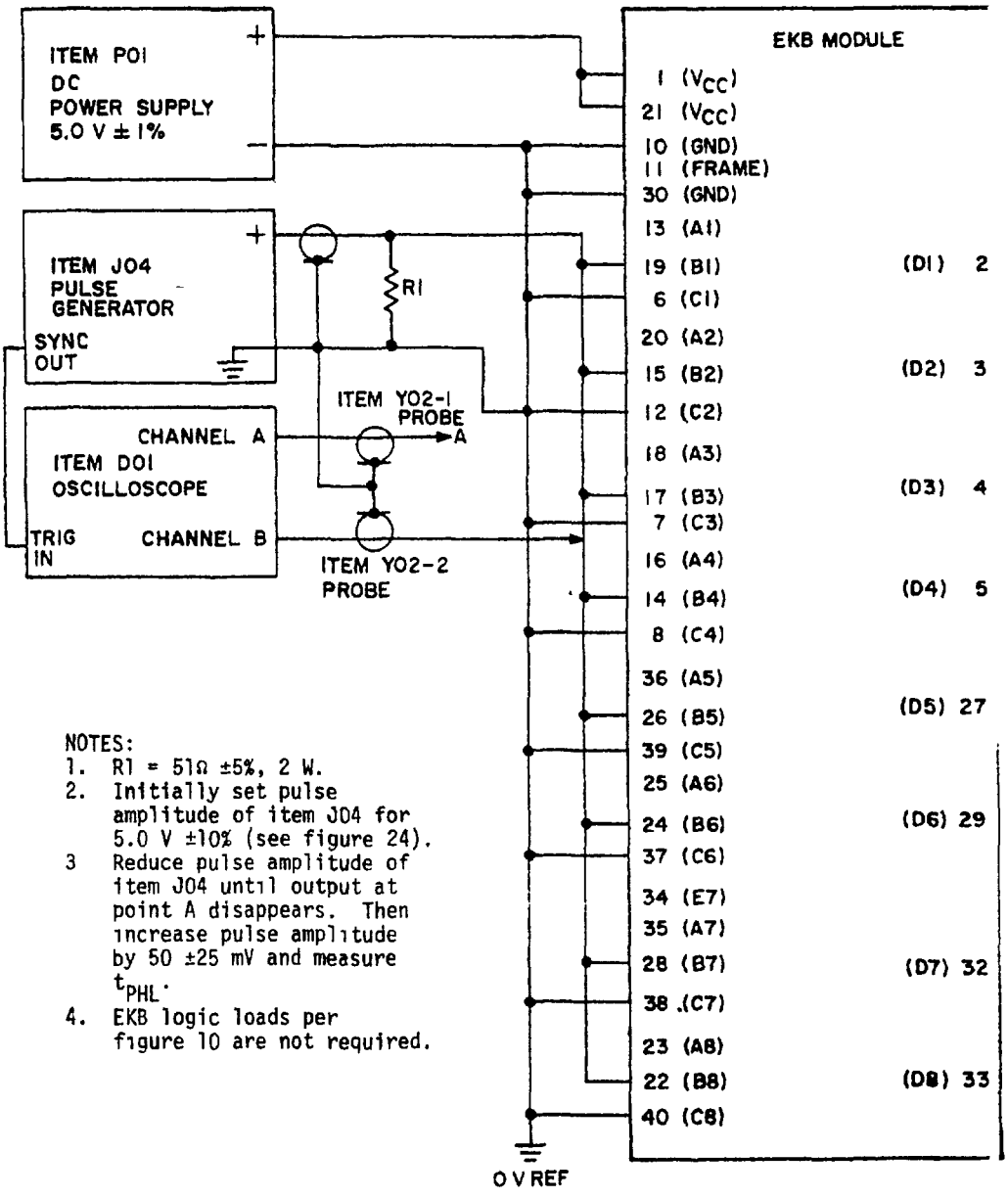
FIGURE 22. Propagation delay time (t_{PLH}), E/C input, 400 Hz, test circuit waveforms for table XVII.

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TABLE XVIII. Schmitt trigger hysteresis (measuring propagation delay time t_{PHL}) check.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
U1	Schmitt trigger hysteresis check (measuring propagation delay time t_{PHL}, C_L = 50 pF)	2	1.9	14.3	ms	23,24
U2		3				
U3		4				
U4		5				
U5		27				
U6		29				
U7		32				
U8	Schmitt trigger hysteresis check (measuring propagation delay time t_{PHL}, C_L = 50 pF)	33	1.9	14.3	ms	23,24

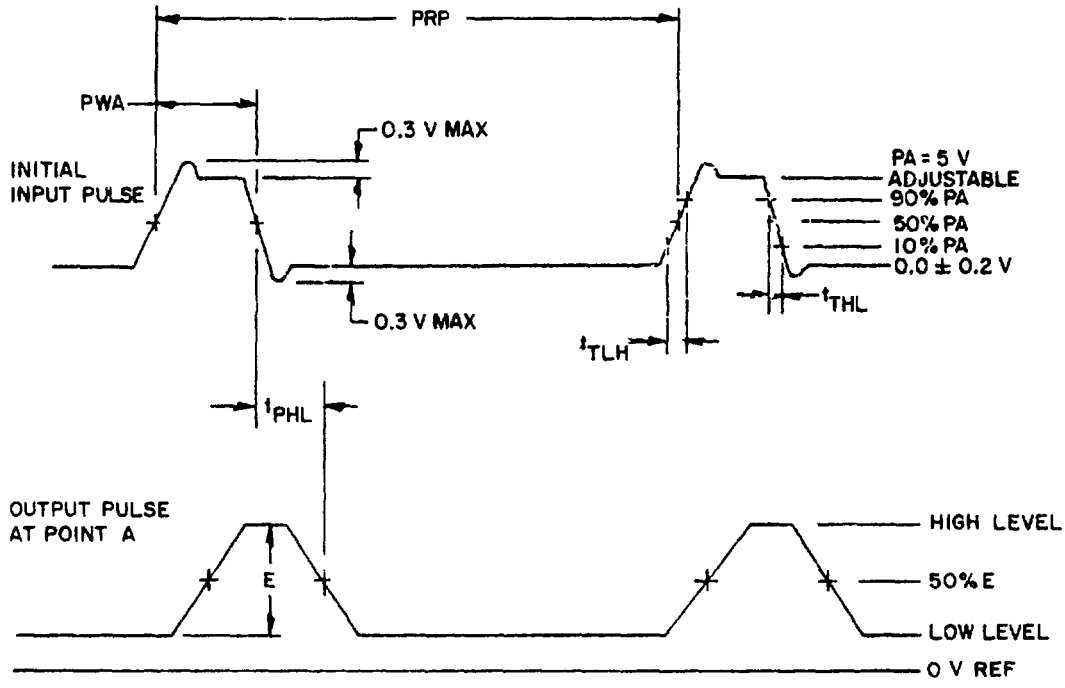
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NOTES:
1. R1 = 51Ω ±5%, 2 W.
2. Initially set pulse amplitude of item J04 for 5.0 V ±10% (see figure 24).
3 Reduce pulse amplitude of item J04 until output at point A disappears. Then increase pulse amplitude by 50 ±25 mV and measure t_{PHL}.
4. EKB logic loads per figure 10 are not required.

FIGURE 23 Schmitt trigger hysteresis (measuring propagation delay time t_{PHL}) check test circuit for table XVIII.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL}$ = 10 ±2 µs.
PWA = 5 ±0.2 ms
PRP = 50 ±5 ms.

FIGURE 24 Schmitt trigger hysteresis (measuring propagation delay t_{PHL}) check test circuit waveforms for table XVIII.

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TABLE XIX. Output crosstalk (low-level and high-level) test.

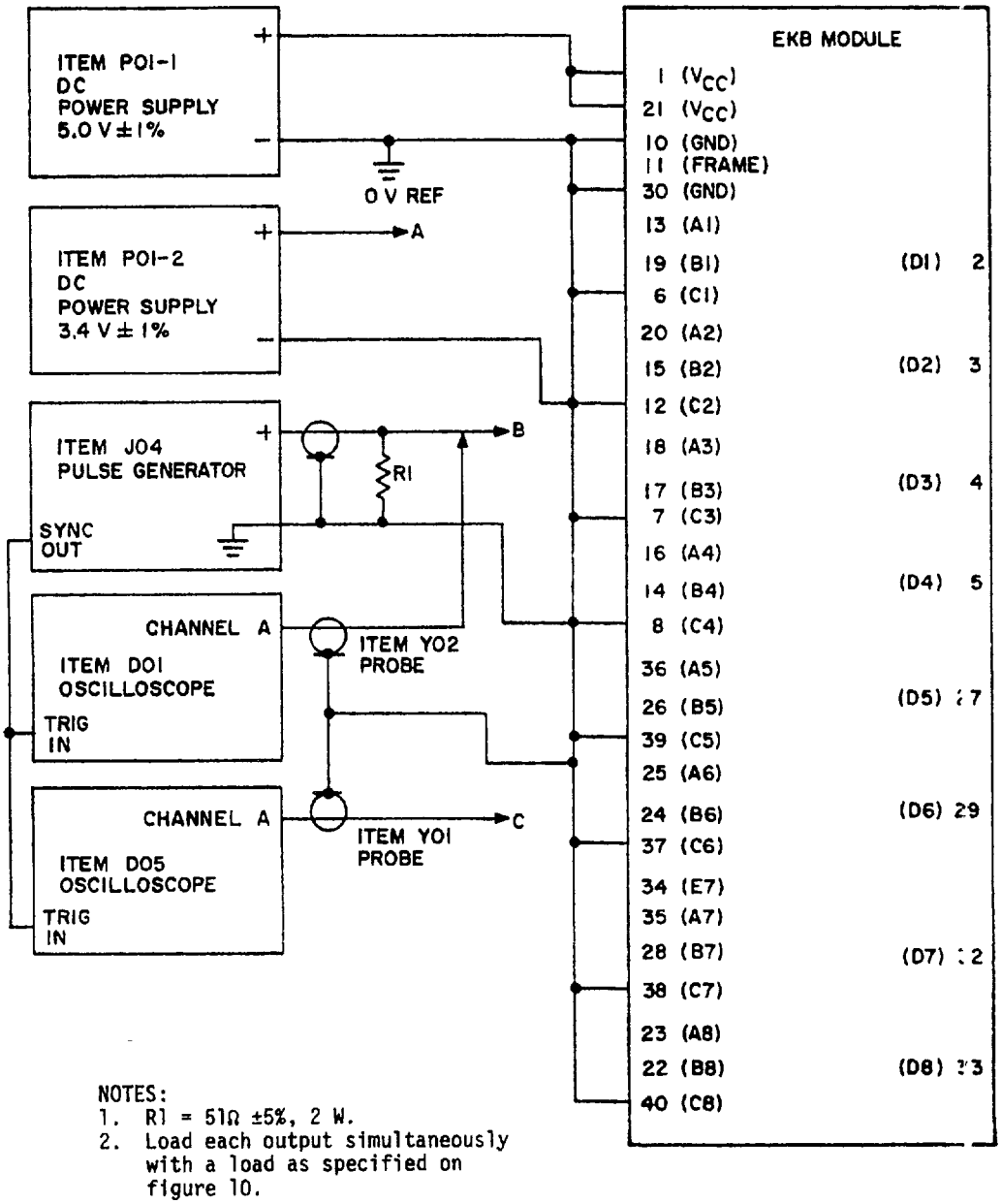
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V1	Output crosstalk, low level, C_L = 50 pF	Open	15,17,14, 26,24,28, 22	2	0	400	mV	25,26
V2			19,17,14, 26,24,28, 22	3				
V3			19,15,14, 26,24,28, 22	4				
V4			19,15,17, 26,24,28, 22	5				
V5			19,15,17, 14,24,28, 22	27				
V6			19,15,17, 14,26,28, 22	29				
V7			19,15,17, 14,26,24, 22	32				
V8	Output crosstalk, low level, C_L =50 pF	Open	19,15,17, 14,26,24, 28	33	0	400	mV	25,26

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TABLE XIX. Output crosstalk (low-level and high-level) test - Continued

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W1	Output crosstalk, high level C_L = 50 pF	19	15,17,14, 26,24,28, 22	2	0	400	mV	25,26
W2		15	19,17,14, 26,24,28, 22	3				
W3		17	19,15,14, 26,24,28, 22	4				
W4		14	19,15,17, 26,24,28, 22	5				
W5		26	19,15,17, 14,24,28, 22	27				
W6		24	19,15,17, 14,26,28, 22	29				
W7		28	19,15,17, 14,26,24, 22	32				
W8	Output crosstalk, high level C_L = 50 pF	22	19,15,17, 14,26,24, 28	33	0	400	mV	25,26

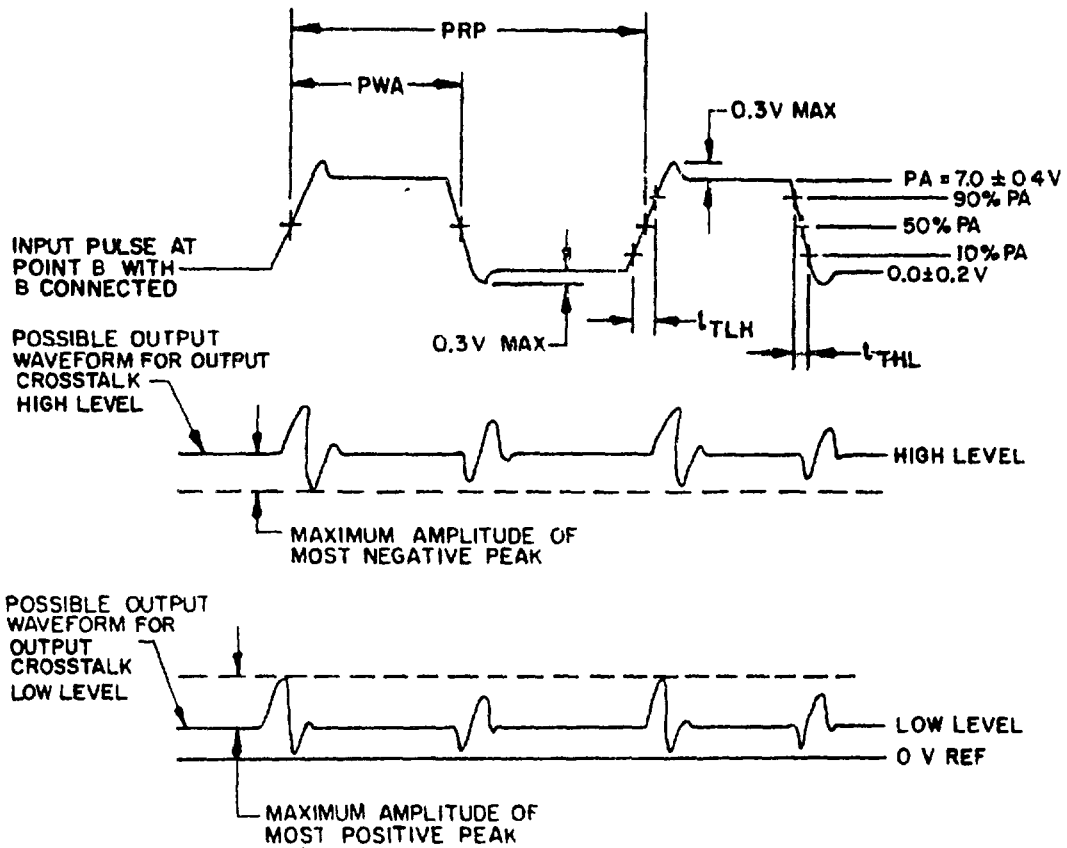
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NOTES:
1. R1 = 51Ω ±5%, 2 W.
2. Load each output simultaneously with a load as specified on figure 10.

FIGURE 25 Output crosstalk (low-level and high-level) test circuit for table XII

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NOTE: Input pulse parameters
$t_{THL} = t_{TLH} = 10 \pm 2$ µs
PWA = 5 ±0.2 ms
PRP = 50 ±5 ms.

FIGURE 26. Output crosstalk (low-level and high-level) test circuit waveforms for table XIX.

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FIGURE XX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
X	Power supply transient amplitude	0	400	mV	27,28

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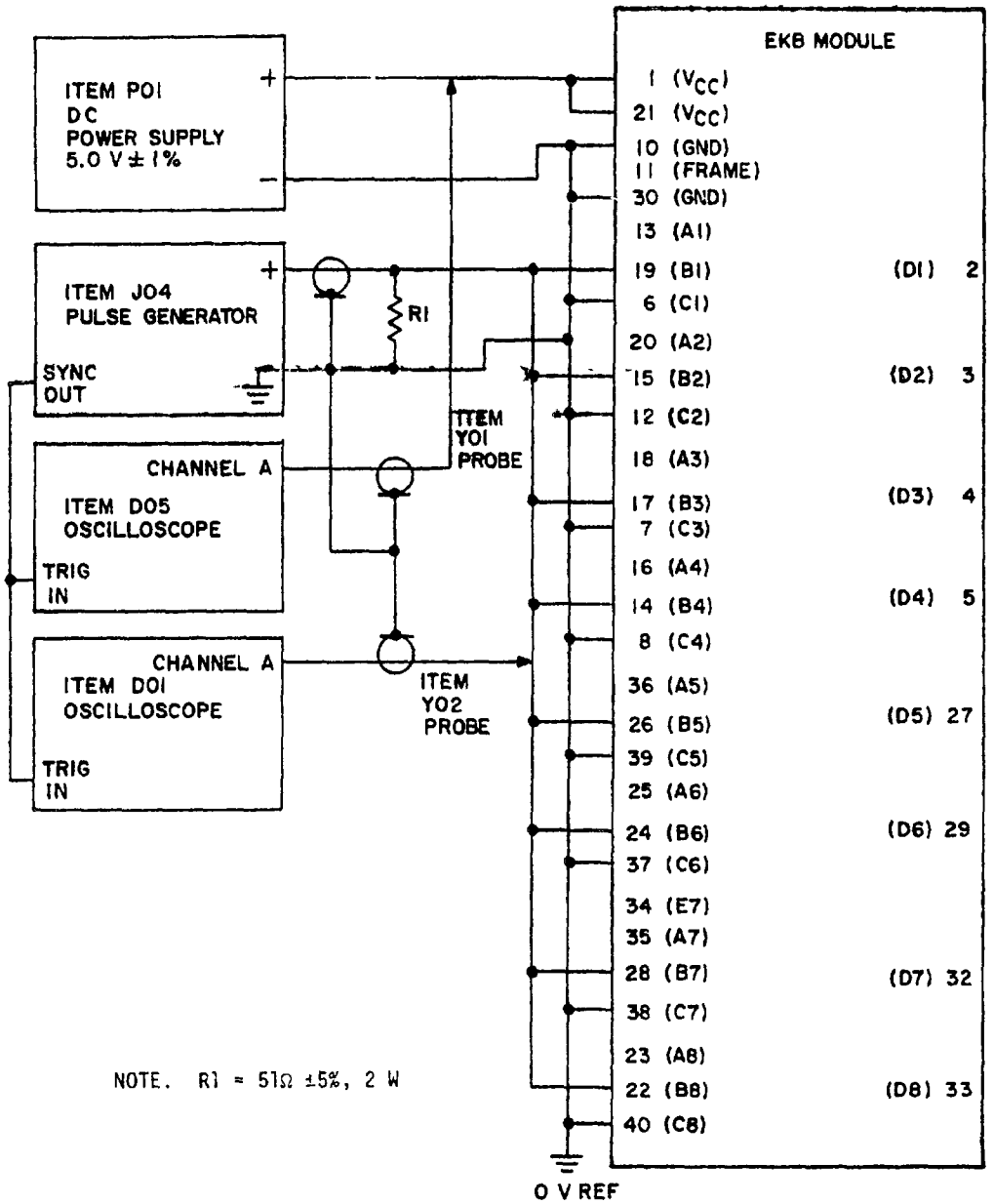
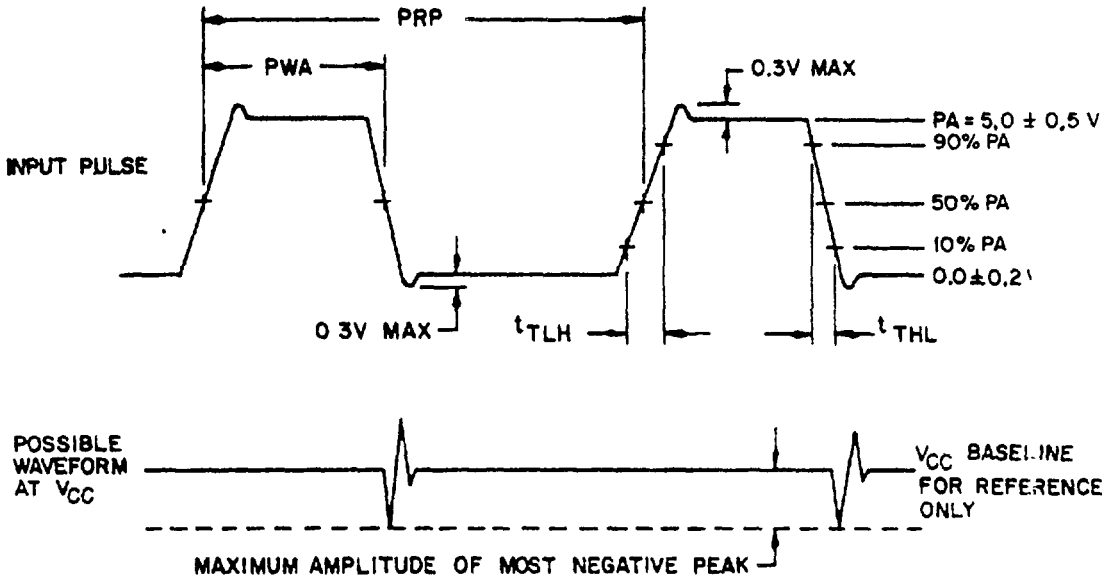


FIGURE 27. Power supply transient amplitude test circuit for table XX.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 10 \pm 2$ µs.
PWA = 5.0 ±0.2 ms.
PRP = 50 ±5 ms

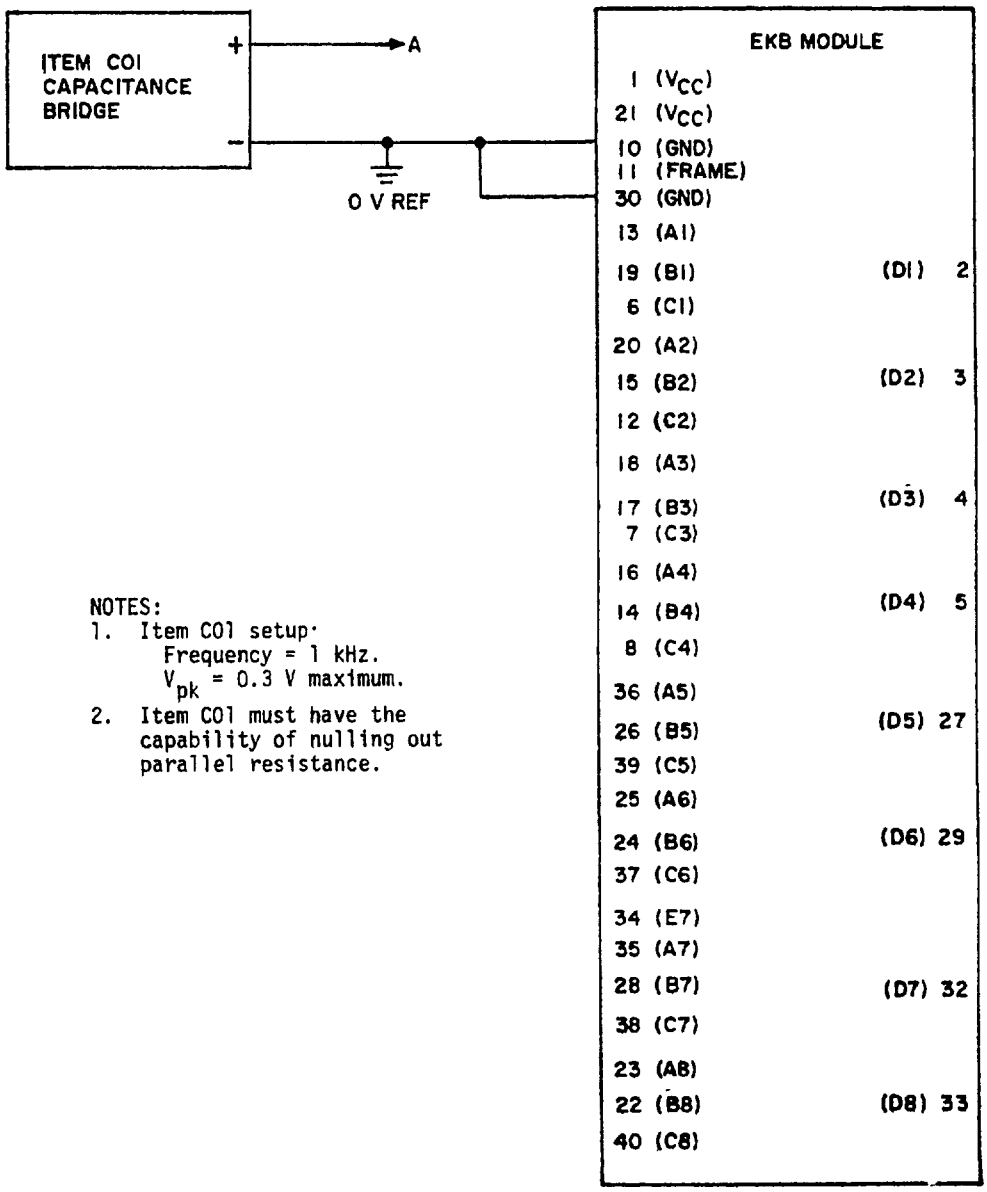
FIGURE 28. Power supply transient amplitude test circuit waveforms for table XI.

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TABLE XXI. Power supply filter capacitance test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
Y1	Power supply filter capacitance, pin 1	1	0.007	0.013	uF	29
Y2	Power supply filter capacitance, pin 21	21	0.007	0.013	uF	29

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NOTES:
1. Item C01 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.
2. Item C01 must have the capability of nulling out parallel resistance.

FIGURE 29. Power supply filter capacitance test circuit for table XXI.

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TABLE XXII. Input short term overload test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
Z	Input short term overload	Must pass all subsequent functional tests - See 4.4.2		-	30, 31

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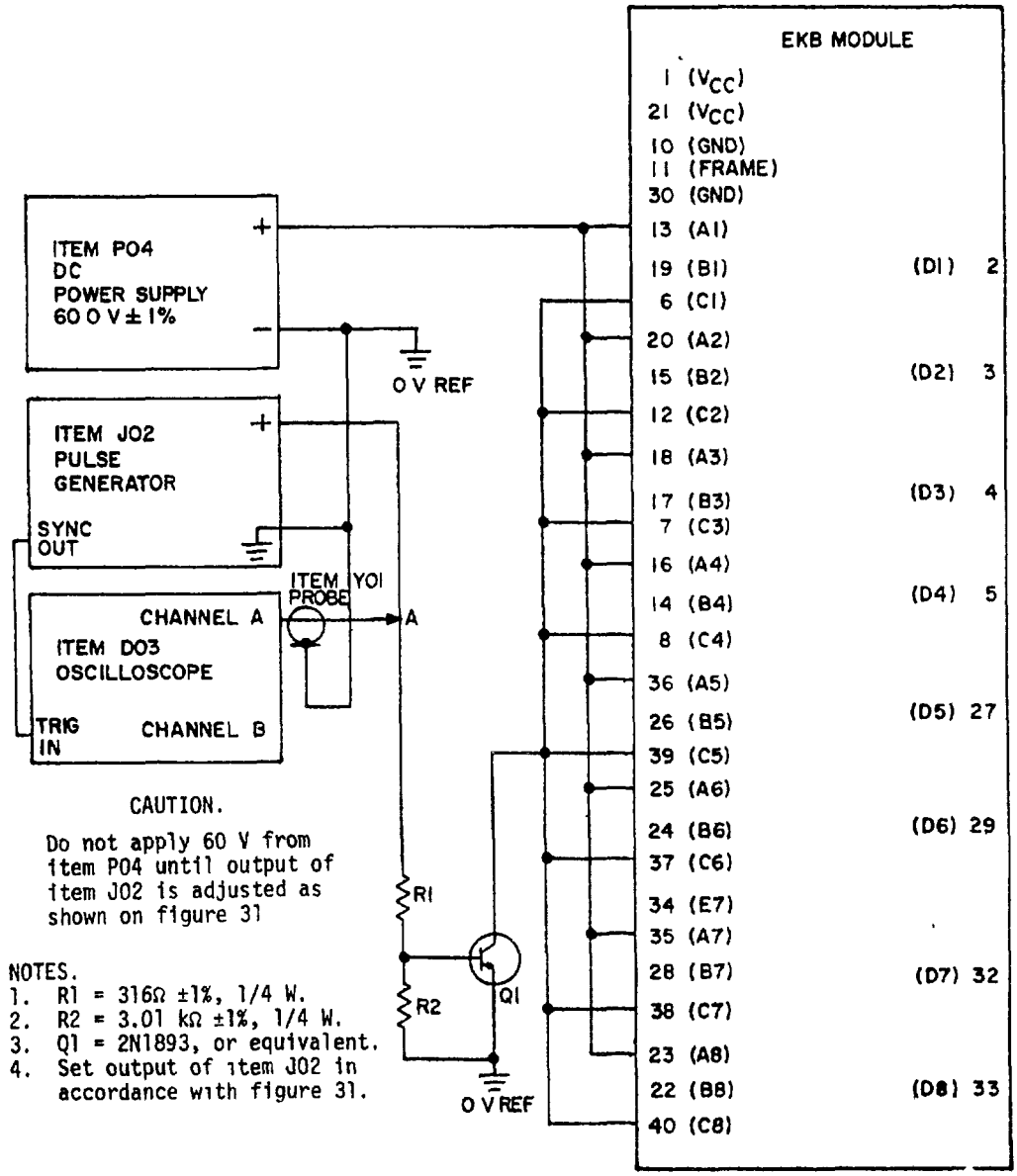
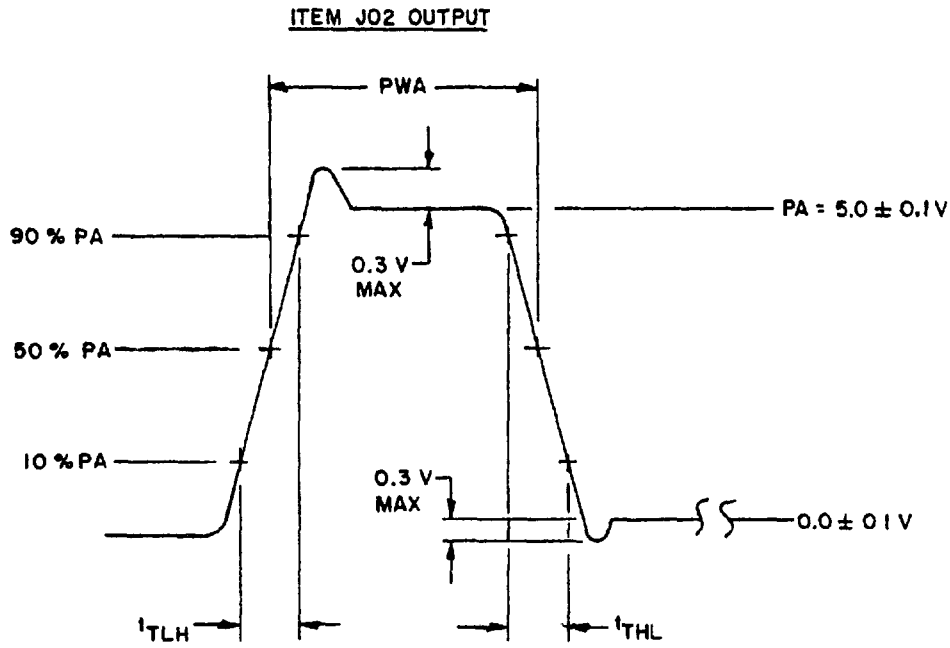


FIGURE 30 Input short term overload test circuit for table XXII

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 10 \pm 10$ ms.
PWA = 1.0 ±0.5 s.
PRP = One shot.

FIGURE 31. Input short term overload test circuit waveform for table XXII.

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